

Modelling Financial Markets

Chapter 9: Realised Volatility



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From Intraday Prices to Variance Forecasts

What we will explain

- Part I, **measurement and diagnosis**: realised variance and its precision, microstructure noise, jumps
- Part II, **modelling and forecasting**
  - stylised facts, the heterogeneous autoregressive (HAR) model, forecast evaluation against GARCH (generalised autoregressive conditional heteroskedasticity, Chapter 5)
  - rough volatility, a machine-learning case study, the research loop assisted by artificial intelligence (AI): extensions that build on the core path, which ends with the model confidence set
- Asymptotic theory, noise- and jump-robust estimators and model extensions sit in the Appendix at the end

What you should be able to do

- compute daily realised variance, its standard error and a confidence interval from intraday prices
- diagnose microstructure noise with a signature plot and test for jumps with false discovery rate (FDR) control
- estimate log-HAR and compare variance forecasts with the quasi-likelihood loss (QLIKE), Diebold–Mariano, Mincer–Zarnowitz and the model confidence set

Volatility Is Latent: Can We Measure It?



Torben G. Andersen, 2012

Photo: Soralum (2012); CC BY-SA 4.0; Wikimedia Commons

- Volatility drives risk (Chapters 7–8), option prices (Chapter 12) and portfolio weights (Chapter 4), but it is never observed
  - One daily return is a single draw from a distribution whose dispersion we want to estimate
  - GARCH (Chapter 5) **infers** today's variance from past daily returns through a model
- Idea: inside one day there are hundreds of price changes; the sum of their squares **measures** the day's variance almost without a model Andersen & Bollerslev (1998)
  - Early idea: Merton (1980) showed that variance, unlike the mean, is estimated better with more frequent data
- Two different objects
  - realised variance: an **ex-post measurement** of day t, known after the close
  - GARCH variance: a **conditional forecast** of day t made with the information of day $t-1$
- Measure first, then model
  - Andersen, Bollerslev, Diebold & Labys (2001), Andersen, Bollerslev, Diebold & Ebens (2001) study the distribution of realised volatility
  - Andersen, Bollerslev, Diebold & Labys (2003) forecast it directly; textbook treatment: Franke, Härdle & Hafner (2019)

Reading Intraday Data

- SPY tracks the Standard & Poor's 500 stock index through an Exchange-Traded Fund (ETF), a fund whose units trade on an exchange
- UTC means Coordinated Universal Time: Bitcoin daily intervals run midnight to midnight in that time zone
- VIX is the Volatility Index of the Chicago Board Options Exchange, an option-price measure of expected volatility
- Up to 18 September 2026; returns $r_i = 100(\ln P_i - \ln P_{i-1})$ in %, P_i: the price at the end of bar i; a bar is labelled by its start time
- Charts and times quoted in this chapter mark the **end** of each 5-minute interval: 09:35 is the first SPY return, 09:30–09:35
- Log returns, not simple percentage changes $100(P_i - P_{i-1})/P_{i-1}$: log returns add up over bars, so the daily return is the sum of the intraday ones; SPY overnight on 7 April 2025: log return -3.27%, simple change -3.22%

Two Markets, Two Clocks: SPY and Bitcoin

Series	Frequency	Period	Days
SPY (ETF)	5-minute bars, 09:30–16:00 New York time	2020–2026	1 489
Bitcoin	5-minute bars, 24-hour days (UTC)	2024–2026	659
SPY, Bitcoin	daily closes	2000/2014–2026	for GARCH
VIX	daily close	2020–2026	implied volatility

- SPY: 78 returns per full day, 43 on the 12 half-days; bars off the 5-minute grid removed; missing bars inside a day carry the last price
- Bitcoin: 288 returns per day; days with fewer than 95% of the bars dropped (88 of 747 calendar days, mostly a data gap in October 2025)

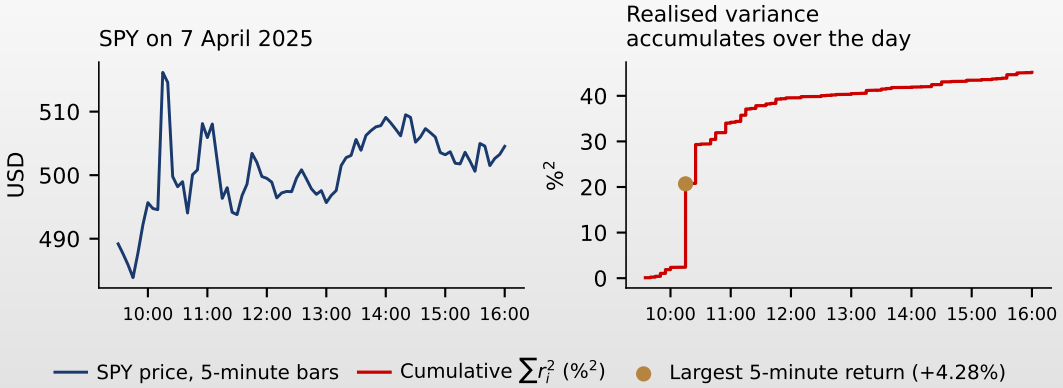
A Model for Intraday Prices

- Notation: day t has M intervals of length $\Delta = 1/M$, returns $r_{t,i}$, daily return $r_t = \sum_i r_{t,i}$
  - Variances in $\%^2$; annualised volatility $\sqrt{252 \times \text{variance}}$ (SPY), $\sqrt{365 \times \text{variance}}$ (Bitcoin); the SPY overnight return (close to next open) is outside the bars
- Log price in continuous time: $dp(s) = \mu(s)\,ds + \sigma(s)\,dW(s)$, with W a Brownian motion (Chapter 11)
  - In words: over an instant ds the log price moves by a drift $\mu(s)\,ds$ (the instantaneous mean return) plus a random shock $\sigma(s)\,dW(s)$, with $dW(s) \sim N(0, ds)$
  - $\sigma(s)$ is the **spot volatility**: it changes within the day and from day to day
- **Integrated variance** of day t: the quantity we want to measure
  - $\mathrm{IV}_t = \int_{t-1}^{t} \sigma^2(s)\,ds$: the spot variance accumulated over the day; in this chapter IV always means integrated variance, never implied volatility
- If σ is independent of W (no leverage) and the drift is zero (or fixed by the path of σ), then given that path the daily return is Normal: $r_t \mid \sigma \sim N(\int \mu, \mathrm{IV}_t)$
  - With leverage (Appendix: Itô semimartingales) this conditional normality fails, but the limit theory of realised variance (RV, next slide) still holds

Definition: Realised Variance

- **Realised variance**: the sum of squared intraday returns
  - $\mathrm{RV}_t = \sum_{i=1}^{M} r_{t,i}^2$, realised volatility $= \sqrt{\mathrm{RV}_t}$
- As $M \to \infty$, $\mathrm{RV}_t \to \mathrm{IV}_t$: the drift vanishes and the sum of squares converges to the **quadratic variation** Barndorff-Nielsen & Shephard (2002)
  - The mean of SPY 5-minute returns is below 1% of their standard deviation: it can be ignored
- No parameters to estimate: RV_t is a **model-free** measurement of one day's variance
- The squared daily return r_t^2 is the special case $M = 1$: unbiased (with zero drift) but extremely noisy

One Day, 78 Returns: 7 April 2025 (1/2)



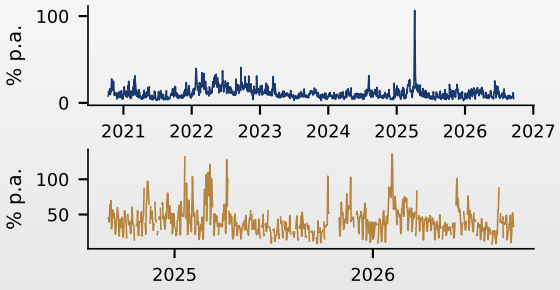
MFM_ch9_intraday_day

One Day, 78 Returns: 7 April 2025 (2/2)

- Left: SPY on 5-minute bars on the most volatile day of the sample; right: running sum of squared returns
- SPY opened with an overnight log return of -3.27%, after tariff announcements; 78 five-minute returns followed
  - $\mathrm{RV} = 45.1\%^2$: annualised realised volatility $\sqrt{252 \times 45.1} = 107\%$
- One 5-minute return of $+4.28\%$ (interval ending 10:15) contributed 41% of the day's realised variance
- Open-to-close the day ended $+3.09\%$: the daily return hides the intraday swings that RV records; by RV it is one of the most turbulent days since 2020

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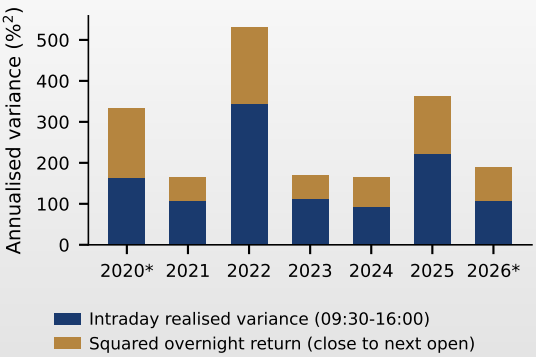
Daily Realised Volatility: SPY and Bitcoin



- Annualised, from 5-minute returns; SPY: regular US session; Bitcoin: 24 hours; the gap in October 2025 is missing data
- SPY: volatility from mean RV $\sqrt{252\,\overline{\mathrm{RV}}} = 12.9\%$ p.a., median day 9.7%, range 2.4% to 106.7%
  - Volatility computed from mean variance exceeds the median: a few very volatile days raise the root-mean-square of daily volatilities
- Bitcoin: $\sqrt{365\,\overline{\mathrm{RV}}} = 45.1\%$ p.a., median 37.8%, maximum 136.1% on 2026-02-05
- Both series show **clustering**, as with GARCH (Chapter 5), but every point is a measurement, not a model output

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Where Is the Risk: Day or Night?



- SPY, mean daily variance by year, annualised: intraday realised variance plus squared overnight return; 2020 and 2026 are partial years
- Overnight returns carry 38% of SPY's daily variance, although the market is closed for 17.5 hours
  - News (earnings, macro data, Asian and European sessions) accumulates and is priced at the open
- Annualised: close-to-close 16.6%; open-to-close 13.6%; overnight 10.1%
- Largest overnight move: -4.08% on 2024-08-05
- For daily risk we use **total** variance: $\mathrm{RV}_t + r^2_{\mathbf{overnight},t}$ (annualised mean 16.4%)
  - Alternatives: scale intraday RV up, or weight the two parts optimally Hansen & Lunde (2005)

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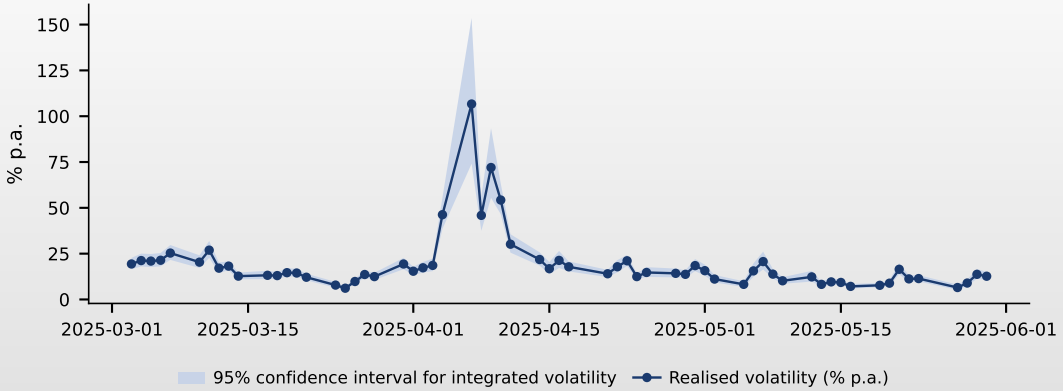
Adding Up Day and Night Variance

- Variances add, volatilities do not: square each annualised volatility first (units $\%^2$ per year)
- SPY, from the slides above; variance calculations use unrounded volatilities
  - intraday RV: 12.9% (slide *Daily Realised Volatility*), variance 167
  - total variance $\mathrm{RV}_t + r^2_{\text{overnight},t}$: 16.4%, variance 270
  - overnight part: $270 - 167 = 103$, i.e. $\sqrt{103} \approx 10.1\%$; share $103/270 = 38\%$
- Why not $13.6^2 + 10.1^2$? The open-to-close 13.6% squares the whole session return, not each 5-minute return
  - $(\sum_i r_{t,i})^2 = \mathrm{RV}_t + 2\sum_{i<j} r_{t,i}\, r_{t,j}$: the cross products are zero only in expectation; in this sample they add about 17 ($13.6^2 \approx 184$)
  - close-to-close (16.6%) also contains the cross term $2\, r_{\text{overnight},t}\, r_{\text{open-close},t}$
- Check: a stock with intraday volatility 20% and overnight volatility 15% has total volatility $\sqrt{400 + 225} = 25\%$ and overnight share $225/625 = 36\%$

How Precise Is Realised Variance?

- Feasible central limit theorem (CLT) for RV Barndorff-Nielsen & Shephard (2002): $\sqrt{M}\,(\mathrm{RV}_t - \mathrm{IV}_t)/\sqrt{2\,\mathrm{RQ}_t} \to N(0,1)$
  - In words: the error $\mathrm{RV}_t - \mathrm{IV}_t$, scaled by $\sqrt{M}$ and by its estimated standard deviation, is approximately standard Normal
  - $\mathrm{IQ}_t = \int_{t-1}^{t} \sigma^4(s)\,ds$: the integrated quarticity, which sets the variance of RV; it is estimated by the realised quarticity $\mathrm{RQ}_t = \frac{M}{3}\sum_i r_{t,i}^4$
- Standard error: $\mathrm{se}(\mathrm{RV}_t) = \sqrt{\frac{2}{3}\sum_i r_{t,i}^4}$
- 95% confidence interval (CI) on the log scale (always positive): $\mathrm{RV}_t \exp\left(\pm 1.96\,\mathrm{se}(\mathrm{RV}_t)/\mathrm{RV}_t\right)$
  - With constant volatility, $\mathrm{se}/\mathrm{RV} \approx \sqrt{2/M}$: 16% for $M = 78$
- The limiting variance $2\,\mathrm{IQ}_t$ is itself random (a mixed Normal limit); replacing it by $2\,\mathrm{RQ}_t$ is legitimate because the convergence is **stable** (formal statement in the Appendix)

Confidence Intervals for Daily Volatility (1/2)



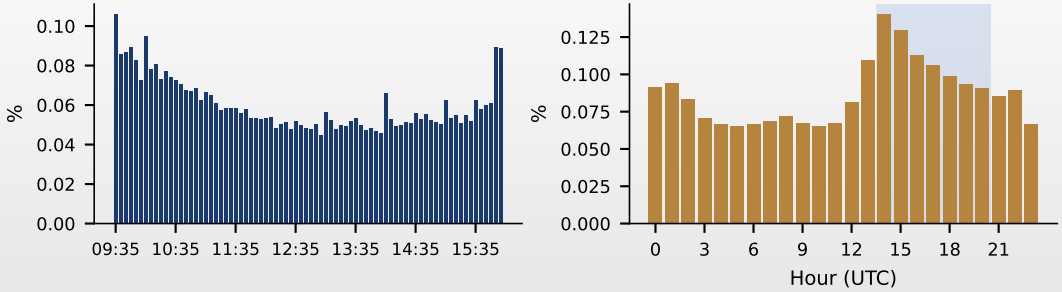
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Confidence Intervals for Daily Volatility (2/2)

- SPY, March–May 2025; band: 95% confidence interval for $\sqrt{252\,\mathrm{IV}_t}$ from realised quarticity
- Median relative width of the interval (for volatility): 37% of its level
  - The band assumes a continuous price and negligible noise: on a jump day RV tends to $\mathrm{IV}_t + \sum_j \kappa_j^2$, and $\sum r^4$ keeps the term κ^4, so the band has no nominal coverage for either the total variation or IV_t (section Jumps)
- 9 April 2025 (the day the tariff pause was announced): volatility 72%, CI [55%, 93%]
  - The interval widens when a few returns dominate the sum: $\sum r^4$ grows faster than $(\sum r^2)^2$
- From variance to volatility: the band takes the square root of the variance endpoints, $\sqrt{\mathrm{RV}_t}\exp\left(\pm 0.98\,\mathrm{se}(\mathrm{RV}_t)/\mathrm{RV}_t\right)$, so the log half-width is halved
  - Toy: $\mathrm{se}/\mathrm{RV} = 0.20$ gives variance factors $\exp(\pm 0.392) = [0.68, 1.48]$ and volatility factors $\exp(\pm 0.196) = [0.82, 1.22]$; annualising multiplies both endpoints by $\sqrt{252}$ and leaves the factors unchanged
  - SPY median $\mathrm{se}/\mathrm{RV} = 0.19$: volatility factors $[0.83, 1.20]$, the 37% median width above and the $\pm 20\%$ rule below
- Still far more precise than r_t^2, whose relative error is of order $\sqrt{2} \approx 141\%$ under the Normal distribution
- Practical rule: a single day's volatility is known to about $\pm 20\%$ (95% band), averages over weeks much more precisely

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Intraday Seasonality (1/2)



SPY: mean |5-minute return| (%), New York time
US equity session (13:30-21:00 UTC, depending on daylight saving)
Bitcoin: mean |5-minute return| (%) by UTC hour

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Intraday Seasonality (2/2)

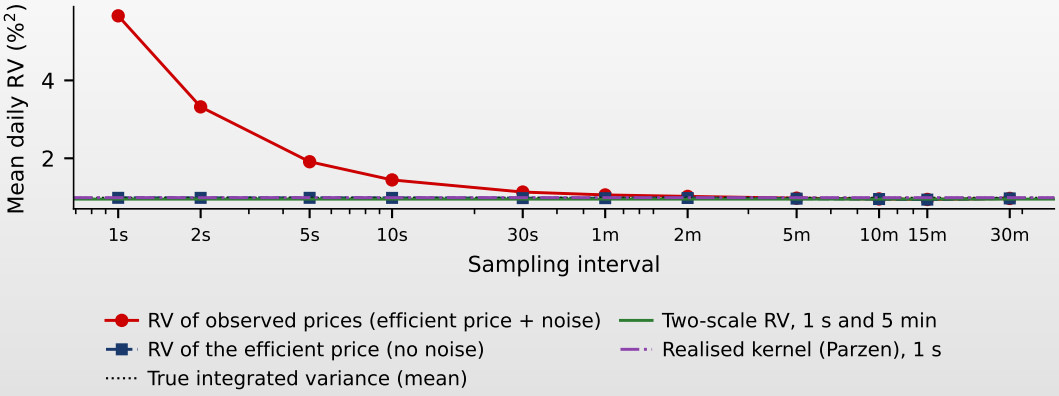
- Left: SPY, mean $|r_{t,i}|$ by 5-minute interval of the day (full days only; ticks mark the end of the interval, New York time); right: Bitcoin by UTC hour
- SPY: a **U-shape**: 0.106% in the first 5 minutes, 0.050% around midday, 0.089% in the last 5 minutes Andersen & Bollerslev (1997)
  - The opening auction incorporates overnight news; the close concentrates index and fund rebalancing
- Bitcoin trades 24/7 but is most volatile at 14:00 UTC, near the US equity opening (09:30 New York: 13:30 UTC in summer, 14:30 UTC in winter); quietest at 10:00 UTC (ratio 2.2)
- RV sums over the day, so seasonality does not bias it
  - it does bias bipower variation (BV, defined in section Jumps) and makes the jump test over-reject
  - the remedy, dividing returns by a periodicity factor f_i Boudt, Croux & Laurent (2011), is applied to the jump test in section Jumps (slide *Many Days, Many Tests*; ▸ Appendix)

MFM_ch9_seasonality

Why Not Use Every Tick?

- Observed prices are not the efficient price: $p_i^{\text{obs}} = p_i^* + \varepsilon_i$, so $r_i = r_i^* + \varepsilon_i - \varepsilon_{i-1}$
  - p_i^*: the efficient (true) log price, ε_i: the noise in the observed price, r_i^*: the efficient return
  - Sources of ε: bid–ask bounce, tick size (price discreteness), stale quotes (Chapter 10)
- With i.i.d. noise of variance ω^2 and n returns per day Zhang, Mykland & Aït-Sahalia (2005), Hansen & Lunde (2006)
  - $\mathbb{E}[\mathrm{RV}_t^{(n)} \mid \sigma] = \mathrm{IV}_t + 2n\omega^2$ (zero drift, no leverage): each squared return carries $2\omega^2$ of noise, so the bias **grows** with the sampling frequency
  - Adjacent returns share ε_{i-1} with opposite signs: $\mathrm{cov}(r_i, r_{i-1}) = -\omega^2$
- Trade-off: $\mathbb{E}[(\mathrm{RV}^{(n)} - \mathrm{IV})^2] \approx 2\,\mathrm{IQ}/n + 4n^2\omega^4$, discretisation error plus squared bias
  - MSE (mean squared error) optimal number of returns $n^* = (\mathrm{IQ}/(4\omega^4))^{1/3}$ Bandi & Russell (2008): more returns cut the first term, but raise the second; derivation in the ▸ Appendix
- Numbers: $\mathrm{IV} = 1\%^2$ with constant volatility within the day, so $\mathrm{IQ} = \mathrm{IV}^2 = 1\ (\%^4)$; $\omega = 0.01\%$, so $\omega^2 = 10^{-4}\%^2$: bias 0.016 at 5 minutes ($n = 78$), 4.680 at 1 second; $n^* = 292$ (one return every 80 s)
  - Check: on a busier day, $\mathrm{IV} = 4\%^2$ (constant volatility, so $\mathrm{IQ} = 16$): $n^* = (16/(4 \times 10^{-8}))^{1/3} \approx 737$, one return every 32 s: more signal tolerates finer sampling

The Volatility Signature Plot: Simulation (1/2)



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The Volatility Signature Plot: Simulation (2/2)

- A signature plot shows mean RV against the sampling interval; here: 100 simulated days, one price per second; efficient price with random daily volatility plus i.i.d. noise, $\omega = 0.01\%$
- True mean IV $= 0.98$; RV of observed prices: 5.66 at 1 second, 1.06 at 1 minute, 0.98 at 5 minutes
  - The bias matches $2n\omega^2$ (slide *Why Not Use Every Tick?*)
- Without noise, RV is flat in the sampling interval: the rise at high frequency is the fingerprint of noise
- Noise-robust estimators on 1-second data: two-scale RV 0.95, realised kernel 0.99
  - error standard deviation across days: 5-minute RV 0.187, two-scale 0.150, kernel 0.045
  - using all data pays off only with a noise correction

MFM_ch9_noise_simulation

Remedies: Sparse Sampling and Subsampling

- **Sparse sampling**: use one return every 5 minutes instead of every tick
  - Simple and robust; throws away most of the data Liu, Patton & Sheppard (2015)
- **Subsampling**: compute RV on K shifted grids (offsets $0, 1, \ldots, K-1$) and average; $\mathrm{RV}_t^{(k)}$: RV on the grid with offset k
  - $\mathrm{RV}_t^{\mathsf{avg}} = \frac{1}{K}\sum_{k=0}^{K-1} \mathrm{RV}_t^{(k)}$: same bias as one sparse grid, smaller variance
- **Two-scale RV** (TSRV) Zhang, Mykland & Aït-Sahalia (2005): remove the remaining bias with the all-data RV
  - $\mathrm{TSRV}_t = \mathrm{RV}_t^{\mathsf{avg}} - \frac{\bar{n}}{n}\mathrm{RV}_t^{\mathsf{all}}$, $\bar{n} = (n - K + 1)/K$; $\mathrm{RV}_t^{\mathsf{all}}$: RV on all n returns, which estimates the noise term $2n\omega^2$; divide by $1 - \bar{n}/n$ in small samples
  - $\bar{n}$ is the average number of returns per sparse grid; our grids also keep the open and the close, so $\bar{n}$ is their actual average count

Realised Kernels

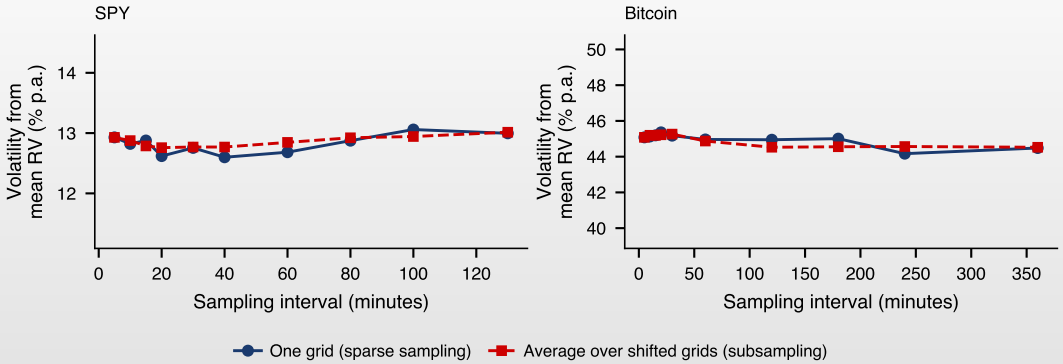


Neil Shephard, 2004

Photo: Renate Schmid (2004); CC BY-SA 2.0 de; Wikimedia Commons

- **Realised kernel** Barndorff-Nielsen, Hansen, Lunde & Shephard (2008): add weighted autocovariances of returns
  - $\mathrm{RK}_t = \gamma_0 + 2\sum_{h=1}^{H} k\left(\frac{h}{H+1}\right)\gamma_h$, $\gamma_h = \sum_i r_{t,i}\, r_{t,i-h}$
  - Parzen weight function k: 1 at 0, falling smoothly to 0 at 1; keeps RK positive
- Noise makes γ_1 negative; the kernel adds it back and cancels the bias
- Bandwidth H grows with the noise-to-signal ratio
  - $H \approx 3.5\,\xi^{4/5} n^{3/5}$, with $\xi^2 = \omega^2/\sqrt{\mathrm{IQ}}$ the noise-to-signal ratio
- On liquid 5-minute data the kernel changes RV only slightly (next slides); it pays off with tick data

The Signature Plot on Real Data (1/2)



MFM_ch9_signature

The Signature Plot on Real Data (2/2)

- Annualised volatility from mean RV by sampling interval: one grid vs the average over all shifted grids
- Flat: SPY 12.9% at 5 minutes, 12.8% at 30, 13.0% at 130; Bitcoin 45.1% at 5 minutes, 44.5% at 6 hours
  - First-order autocorrelation of 5-minute returns: -0.008 (SPY), 0.005 (Bitcoin): no visible noise at 5 minutes
- Limit of this dataset: 5-minute bars can diagnose 5-minute and coarser sampling only; they say nothing about noise below 5 minutes
- On 5-minute SPY data: two-scale RV / RV $= 0.97$, realised kernel / RV $= 0.95$; correlations with RV 0.89 and 0.85
- Consistent with Liu, Patton & Sheppard (2015): almost 400 estimators on 31 assets rarely beat 5-minute RV
  - noise-robust estimators pay off when noise is large relative to the signal (tick or second-level data, illiquid assets)

MFM_ch9_signature

Prices Jump: Bipower Variation



Ole E. Barndorff-Nielsen, 2007

Photo: Thomas Steiner (2007); CC BY-SA 2.5; Wikimedia Commons

- Add jumps to the model: $dp = \mu\, ds + \sigma\, dW + dJ$, with J a jump process (news, Chapter 11)
  - Quadratic variation now has two parts: $\mathrm{RV}_t \to \mathrm{IV}_t + \sum_j \kappa_j^2$, with κ_j the jump sizes in day t
  - Why separate them? Jumps are less persistent than diffusive volatility, and they price differently (options, Chapter 12)
- **Bipower variation** Barndorff-Nielsen & Shephard (2004): products of adjacent absolute returns
  - $\mathrm{BV}_t = \mu_1^{-2} \frac{M}{M-1} \sum_{i=2}^{M} |r_{t,i}|\,|r_{t,i-1}| \to \mathrm{IV}_t$
  - $\mu_1 = \mathbb{E}|Z| = \sqrt{2/\pi} = 0.798$ for $Z \sim N(0,1)$, so $\mu_1^{-2} = \pi/2 = 1.571$; $\frac{M}{M-1}$: finite-sample normalisation
- A jump enters one return, which is multiplied by a small neighbour: its effect on BV vanishes as M grows; the jump part of the day is $\max(\mathrm{RV}_t - \mathrm{BV}_t, 0)$
  - Example, $M = 4$ returns $0.1, 0.1, 0.1, 1.0$ (%): $\mathrm{RV} = 1.03$, $\mathrm{BV} = \frac{\pi}{2} \cdot \frac{4}{3}(0.01 + 0.01 + 0.10) = 0.25$, jump part 0.78 (%^{2})

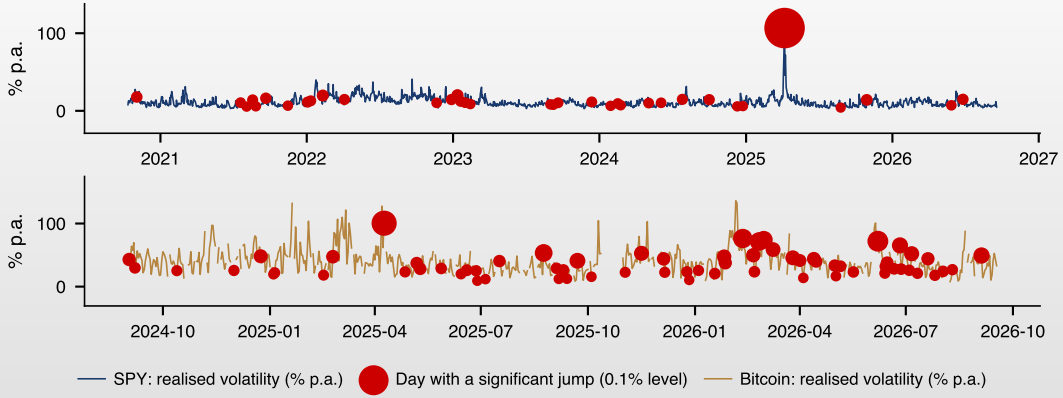
A Test for Jumps

- Ratio statistic Huang & Tauchen (2005), Barndorff-Nielsen & Shephard (2006):

$$z_t = \frac{(\mathrm{RV}_t - \mathrm{BV}_t)/\mathrm{RV}_t}{\sqrt{\theta\, M^{-1} \max\left(1, \mathrm{TQ}_t/\mathrm{BV}_t^2\right)}}$$

  - $\theta = \pi^2/4 + \pi - 5 = 0.609$; tripower quarticity $\mathrm{TQ}_t = M\mu_{4/3}^{-3} \frac{M}{M-2} \sum_i |r_{t,i}|^{4/3} |r_{t,i-1}|^{4/3} |r_{t,i-2}|^{4/3}$, $\mu_{4/3} = \mathbb{E}|Z|^{4/3}$
  - In words: the relative jump share $(\mathrm{RV}_t - \mathrm{BV}_t)/\mathrm{RV}_t$ divided by its standard error; θ: the asymptotic variance constant; TQ estimates IQ_t robustly to jumps
- Under no jumps $z_t \to N(0,1)$; a jump makes $\mathrm{RV} > \mathrm{BV}$ and z_t large and positive
- One test per day: at level α we expect $\alpha\times$ (number of days without a jump) false alarms, at most $\alpha\times$ days
  - We use $\alpha = 0.1\%$ (critical value 3.09) for the jump days plotted on the next slide
- Significant day: jump part $J_t = \mathrm{RV}_t - \mathrm{BV}_t$, continuous part $C_t = \mathrm{RV}_t - J_t$; otherwise $J_t = 0$

Jump Days: SPY and Bitcoin (1/2)



MFM_ch9_jumps

Jump Days: SPY and Bitcoin (2/2)

- Red points: days with a significant jump at the 0.1% level; point size proportional to the jump part J_t
- SPY: 35 jump days of 1 489 (2.4%); jumps are 2.8% of total realised variance
  - on a jump day the jump is on average 35% of that day's RV
  - BV/RV averages 0.95 over all days
  - largest statistic: 2021-08-27 ($z = 5.2$), 10:00–10:05: the Jackson Hole speech of the Fed chair
- Bitcoin: 67 jump days of 659 (10.2%), but only 1.8% of variance
  - Many small jumps on quiet days (weekends): one isolated move dominates a calm day
  - these counts are the replication check for Seminar 9, B4
- The turbulent days of April 2025 are mostly **not** jump days: many large returns in a row raise BV as well

MFM_ch9_jumps

Many Days, Many Tests: Spurious Jumps

- 1 489 daily tests at 5%: 235 rejections against 74 expected under no jumps; which of them are false?
- Part of the excess is intraday seasonality (section Realised Variance): dividing returns by the periodicity factor f_i Boudt, Croux & Laurent (2011) already cuts the 5% rejections from 235 to 167
- Benjamini & Hochberg (1995): sort the p-values and reject the k smallest
  - k: the largest index with $p_{(k)} \leq kq/N$; $p_{(k)}$: k-th smallest p-value, N: number of tests, q: target FDR
  - here each day's p-value is the one-sided $1 - \Phi(z_t)$ of the ratio statistic
  - the FDR, $\mathbb{E}$[false/rejections], stays $\leq q$ (false/rejections $:= 0$ when nothing is rejected)
  - The guarantee needs valid p-values that are independent or positively dependent Benjamini & Yekutieli (2001); daily jump tests are only approximately so

The Benjamini–Hochberg Rule on Five Tests

- Five days, sorted p-values $0.003, 0.012, 0.024, 0.045, 0.60$; target $q = 0.05$, $N = 5$
- Thresholds kq/N for $k = 1, \ldots, 5$: $0.01, 0.02, 0.03, 0.04, 0.05$
- Compare position by position
  - $0.003 \leq 0.01$, $0.012 \leq 0.02$, $0.024 \leq 0.03$: pass; $0.045 > 0.04$, $0.60 > 0.05$: fail
  - largest passing index $k = 3$: reject the three smallest p-values
- Three rules, three answers on the same data
  - no correction ($p \leq 0.05$): 4 rejections; BH at $q = 5\%$: 3; Bonferroni ($p \leq 0.05/5 = 0.01$): 1
- Check (step-up): with p-values $0.003, 0.012, 0.032, 0.038, 0.60$, the third fails its own threshold ($0.032 > 0.03$) but the fourth passes ($0.038 \leq 0.04$), so $k = 4$ and four days are rejected

Jump Counts under Multiplicity Control

- SPY, 1 489 days
  - FDR at 5%: 42 jump days; at 1%: 9; Bonferroni (family-wise error rate, FWER, 5%): 7; FDR 5% after the periodicity correction: 17
  - from 235 unadjusted rejections to 17: most rejections do not survive the two corrections for seasonality and multiplicity
- Bajgrowicz, Scaillet & Treccani (2016): without multiplicity control most detected jumps are spurious; the true ones are rare and cluster on news
- Christensen, Oomen & Podolskij (2014): at tick frequency most 5-minute "jumps" are steep but continuous bursts of volatility (fact or friction)
- A daily test answers "did day t contain a jump?"; locating the jump within the day needs an intraday test (Lee–Mykland, ▸ Appendix)

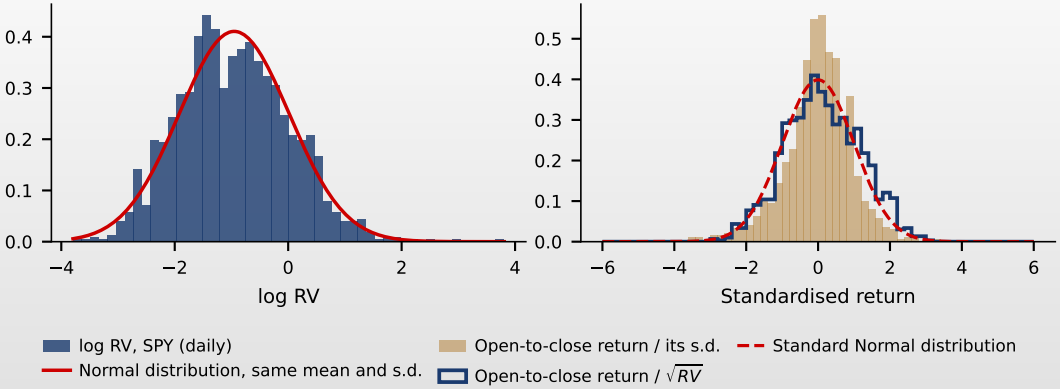
Check Yourself: Measuring Daily Variance

- Why does RV converge to integrated variance as M grows?
  - see slide *Definition: Realised Variance*
- Why does the standard error of RV use fourth powers of returns?
  - see slide *How Precise Is Realised Variance?*
- Why is a 95% band for volatility narrower, on the log scale, than the band for variance?
  - see slide *Confidence Intervals for Daily Volatility*
- Why is SPY's daily risk understated if you use intraday RV alone?
  - see slides *Where Is the Risk: Day or Night?* and *Adding Up Day and Night Variance*

Check Yourself: Noise and Jumps

- Why does sampling every tick overstate variance?
  - see slide *Why Not Use Every Tick?*
- What does a flat signature plot tell you? What does it not tell you?
  - see slide *The Signature Plot on Real Data* (the limit of 5-minute bars)
- Why does bipower variation stay close to IV on a jump day while RV does not?
  - see slide *Prices Jump: Bipower Variation* (the four-return example)
- With 1 489 daily jump tests, why is "$p < 0.05$" not evidence of a jump? What does the FDR control?
  - see slides *Many Days, Many Tests* and *The Benjamini–Hochberg Rule on Five Tests*

Distributions: log RV and Standardised Returns (1/2)



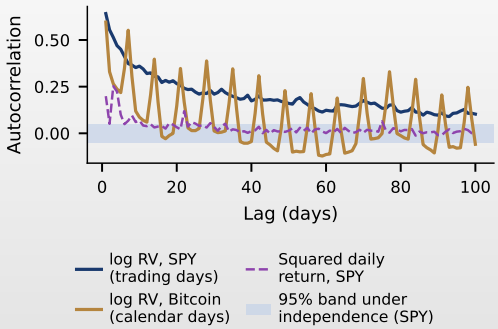
MFM_ch9_distributions

Distributions: log RV and Standardised Returns (2/2)

- Left: daily log RV of SPY and the Normal density; right: open-to-close returns scaled by their standard deviation (filled) and by $\sqrt{\mathrm{RV}_t}$ (outline)
- SPY RV: skewness 20.0, kurtosis 550; log RV: skewness 0.29, kurtosis 3.20 (the Normal distribution: 0 and 3)
  - Bitcoin log RV: skewness -0.27, kurtosis 3.33: model volatility in logs
- Open-to-close returns: kurtosis 15.73; divided by $\sqrt{\mathrm{RV}_t}$: 2.65, standard deviation 1.05
  - Bitcoin: from 7.57 to 2.77; both pairs are the replication check for Seminar 9, B5
- Heavy tails of daily returns (Chapter 1) come mostly, ex post, from changing volatility: approximately a **mixture** of Normal distributions with variances IV_t, exact only without leverage and jumps (section Realised Variance)

MFM_ch9_distributions

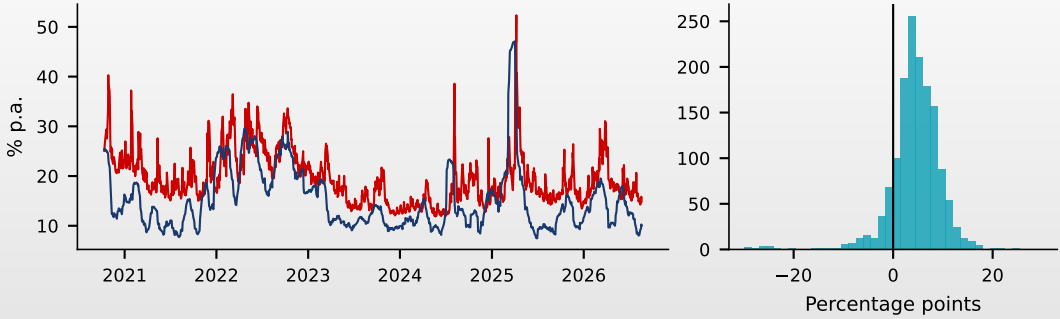
Persistence of Realised Volatility



- Autocorrelations up to 100 days: log RV of SPY (total variance) and Bitcoin, squared daily SPY returns
- SPY log RV: 0.64 at lag 1, 0.24 at one month, still 0.10 at 100 days: **highly persistent** dependence
  - an autoregressive AR(1) model would decay geometrically: $0.64^{22} \approx 0.00005$
  - a sample autocorrelation function (ACF) is consistent with long memory but cannot prove it
- Squared daily returns show the same persistence, hidden by noise: 0.20 at lag 1
- Bitcoin: peaks at 7, 14, 21, ... days: a **weekly cycle**
  - weekend RV is only 0.41 of weekday RV (Saturday 27% vs Wednesday 48% p.a.)
  - no equity or macro news, fewer institutional traders

MFM_ch9_acf

Implied vs Realised Volatility (1/2)



MFM_ch9_vix

Implied vs Realised Volatility (2/2)

- Left: VIX and SPY realised volatility over the next 21 trading days; right: the gap between them
- VIX: 30-day volatility implied by S&P 500 options (Chapter 12)
  - mean 19.5% vs 15.2% realised afterwards
  - VIX was higher on 88% of 1 468 days, by 4.4 percentage points on average
- **Variance risk premium** (VRP) Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009)
  - $\mathrm{VRP}_t = \mathbb{E}_t^{\mathbb{Q}}[\mathrm{RV}_{t,t+\mathbf{21}}] - \mathbb{E}_t^{\mathbb{P}}[\mathrm{RV}_{t,t+\mathbf{21}}]$: risk-neutral minus physical expectation
  - $\mathrm{RV}_{t,t+\mathbf{21}}$: realised variance over the next 21 trading days; $\mathrm{VIX}^{\mathbf{2}}$ approximates the first term
  - the chart shows an **ex-post** spread, a noisy proxy of this ex-ante premium
  - negative spread (March–April 2025): a forecast error, not a negative premium
- As a forecast: regress $\frac{\mathbf{252}}{\mathbf{21}}\mathrm{RV}_{t,t+\mathbf{21}}$ (total variance, annualised $\%^{\mathbf{2}}$) on $\mathrm{VIX}_t^{\mathbf{2}}$ (annualised $\%^{\mathbf{2}}$; 30 calendar days $\approx$ 21 trading days)
  - slope 0.56 (standard error, SE, 0.05) < 1: $\mathrm{VIX}^{\mathbf{2}}$ moves more than the variance that follows, so VIX over-reacts (partly because the premium itself rises in turbulent periods); $R^{\mathbf{2}} = 0.23$: VIX is informative
  - level bias is a separate question, answered by the mean gap of 4.4 percentage points: VIX is on average above realised volatility (the premium); the same slope-versus-level distinction returns in the Mincer–Zarnowitz test

MFM_ch9_vix

The HAR-RV Model

- Corsi (2009): $\mathrm{RV}_{t+\mathbf{1}} = \beta_{\mathbf{0}} + \beta_d \mathrm{RV}_t + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + \varepsilon_{t+\mathbf{1}}$
  - $\mathrm{RV}_t^{(w)} = \frac{\mathbf{1}}{\mathbf{5}} \sum_{j=\mathbf{0}}^{\mathbf{4}} \mathrm{RV}_{t-j}$ (week), $\mathrm{RV}_t^{(m)} = \frac{\mathbf{1}}{\mathbf{22}} \sum_{j=\mathbf{0}}^{\mathbf{21}} \mathrm{RV}_{t-j}$ (month)
  - $\beta_{\mathbf{0}}$: constant; β_d, β_w, β_m: weights of yesterday's RV and of last week's and last month's average RV; $\varepsilon_{t+\mathbf{1}}$: forecast error
  - Bitcoin trades every day: week = 7 days, month = 30 days
- Heterogeneous market hypothesis Müller et al. (1997) (the "H" of HAR; HAR-RV: HAR model of realised variance): daily, weekly and monthly traders; a few components with different persistence mimic long memory
- An AR(22) with only 3 free slopes: weight $\beta_d + \beta_w/5 + \beta_m/22$ on lag 1, $\beta_w/5 + \beta_m/22$ on lags 2–5, $\beta_m/22$ on lags 6–22
- Estimation by ordinary least squares (OLS); errors are heteroskedastic and autocorrelated: use Newey–West standard errors (HAC: heteroskedasticity- and autocorrelation-consistent)
  - log-HAR: the same regression for $\ln \mathrm{RV}$, with the weekly and monthly terms as averages of $\ln \mathrm{RV}$; forecast $\exp(\hat{y} + \hat{s}^{\mathbf{2}}/2)$, $\hat{y}$ the fitted $\ln \mathrm{RV}$, $\hat{s}^{\mathbf{2}}$ the residual variance: the conditional mean if the errors are $N(0, s^{\mathbf{2}})$ (lognormal approximation)
  - For non-Normal errors use the smearing factor $\frac{\mathbf{1}}{T} \sum_t e^{\hat{\varepsilon}_t}$ Duan (1983); with heteroskedastic errors both are approximations

Averaging Logs Changes the Forecast Input

- Consider two daily variances, 1 and 4, in the same units. Their mean log is $(\ln 1 + \ln 4)/2 = \ln 2$; the log of their mean is $\ln(2.5)$. These are different regressors.
- The log-HAR averages daily log variances over its weekly and monthly windows. Exponentiating such an average gives a geometric mean, not an arithmetic mean.
- At forecast date t, every averaging window ends at t. The response is the next day's log variance; including day $t+1$ in an input would reveal part of the answer.
- Exponentiating the predicted log value gives a median under a Normal log error. The factor $\exp(s^2/2)$ targets the mean under the stated error assumption; it is not a change in the HAR coefficients.

HAR Estimates: SPY and Bitcoin

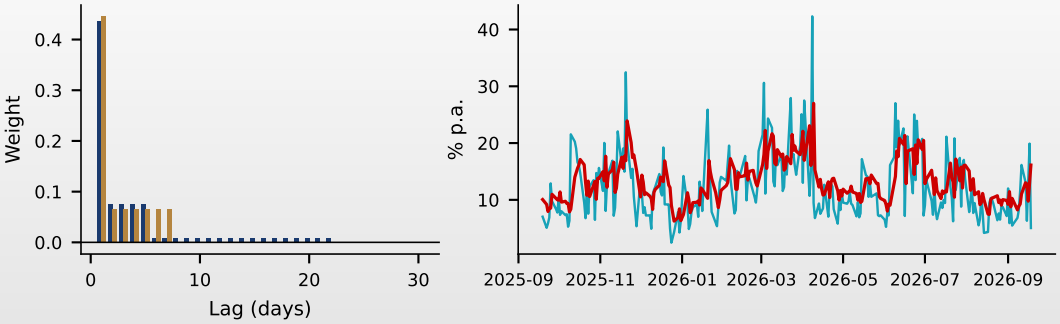
	β_0	β_d	β_w	β_m	R^2
SPY, HAR (levels)	0.273	0.399	0.222	0.114	0.31
	(0.064)	(0.089)	(0.096)	(0.053)	
SPY, log-HAR	−0.079	0.362	0.337	0.167	0.47
	(0.027)	(0.037)	(0.048)	(0.044)	
Bitcoin, HAR (levels)	1.496	0.412	0.409	−0.065	0.34
	(0.412)	(0.089)	(0.122)	(0.097)	
Bitcoin, log-HAR	0.223	0.380	0.465	−0.022	0.36
	(0.094)	(0.041)	(0.080)	(0.098)	

- Full samples; SPY: total daily variance (intraday RV plus squared overnight return), 1 467 days; Bitcoin: 24-hour RV, 547 days on which all three HAR regressors are available
- Newey–West standard errors in brackets

Reading the HAR Estimates

- SPY: all three components are significant; in logs $\hat{\beta}_d = 0.362$, $\hat{\beta}_w = 0.337$, $\hat{\beta}_m = 0.167$
  - Persistence $\hat{\beta}_d + \hat{\beta}_w + \hat{\beta}_m = 0.87$: shocks fade slowly, as in GARCH (Chapter 5)
- $R^2 = 0.47$ for log variance vs 0.31 for variance in levels: not comparable, the dependent variables differ; the out-of-sample losses on a common target (section Forecasting: HAR vs GARCH) decide
- Bitcoin: daily and weekly components matter; the monthly coefficient is close to 0 (-0.022, SE 0.098)
  - Short sample (two years) and a strong weekly cycle: the week absorbs most of the memory

Implied Lag Weights and In-Sample Fit (1/2)



MFM_ch9_har

Implied Lag Weights and In-Sample Fit (2/2)

- Left: weights of log-HAR on each past day (Bitcoin: complete 7- and 30-day windows); right: SPY, last 250 days, realised volatility and log-HAR fitted values
- SPY: yesterday gets 0.437, each of days 2–5 gets 0.075, each of days 6–22 gets 0.008
- Where the numbers come from (the AR(22) weights given with the model definition): lag 1 gets $\hat{\beta}_d + \hat{\beta}_w/5 + \hat{\beta}_m/22 = 0.362 + 0.067 + 0.008 = 0.437$; lags 2–5 get $0.067 + 0.008 = 0.075$
  - The step-shaped weights decay slowly: the monthly component adds $\hat{\beta}_m/22 = 0.0076$ to each of lags 1–22, so $22 \times 0.0076 \approx 0.167 = \hat{\beta}_m$; days 6–22 alone carry $17 \times 0.0076 = 0.129$
  - Three coefficients mimic the slow decay of the autocorrelation function
- Bitcoin: yesterday 0.446, days 2–7 about 0.066 each, almost nothing beyond a week
- In-sample fitted values follow the level of volatility, not the day-to-day noise
- HAR is an AR(22) with 19 restrictions ($\phi_2 = \ldots = \phi_5$, $\phi_6 = \ldots = \phi_{22}$); Newey–West Wald test on SPY: $W = 18.1$, $p = 0.51$, not rejected (derivation in the Appendix)

MFM_ch9_har

Extensions of HAR

- **HAR-CJ** Andersen, Bollerslev & Diebold (2007): separate continuous (C) and jump (J) parts as regressors
  - $\ln \mathrm{RV}_{t+1} = \beta_0 + \beta_{Cd} \ln C_t + \beta_{Cw} \ln C_t^{(w)} + \beta_{Cm} \ln C_t^{(m)}$
  - $\quad + \beta_{Jd} \ln(1 + J_t) + \beta_{Jw} \ln(1 + J_t^{(w)}) + \beta_{Jm} \ln(1 + J_t^{(m)}) + \varepsilon_{t+1}$
  - $C_t^{(w)}, J_t^{(w)}$ ($C_t^{(m)}, J_t^{(m)}$): arithmetic means over the last 5 (22) days, as $\mathrm{RV}_t^{(w)}$; logs are taken after averaging
- **HARQ** (HAR with realised quarticity, RQ) Bollerslev, Patton & Quaedvlieg (2016): less weight on yesterday's RV when it is imprecise
  - measurement error in RV_t attenuates the daily slope; imprecise = high realised quarticity (Appendix)
  - β_d becomes $\beta_d + \beta_Q \sqrt{\mathrm{RQ}_t}$ with $\beta_Q < 0$
- Weekend or holiday dummies, leverage (negative returns raise future RV), implied volatility as an extra regressor
- Multi-step: regress $\sum_{j=1}^{h} \mathrm{RV}_{t+j}$ directly on the same regressors

Which Model Forecasts Tomorrow's Variance Best?

- Target: tomorrow's daily variance; SPY: total variance, 3 January 2022 to 18 September 2026 (1 180 days); Bitcoin: 1 March 2025 onwards (422 days)
  - Every forecast uses only data available the evening before
- Five models
  - log-HAR and HAR on realised variance, re-estimated daily on an expanding window
  - GARCH(1,1) with Student-t errors on daily returns since 2000 (SPY) or 2014 (Bitcoin), re-estimated every 21 days Bollerslev (1986)
  - EWMA (exponentially weighted moving average) with decay factor $\lambda = 0.94$ (Chapter 5) and "yesterday's RV" (random walk)
- HAR sees intraday data; GARCH and EWMA see only daily returns

Evaluating Against a Noisy Target

- True variance is unobserved; we compare forecasts F_t with the proxy RV_t
  - For SPY the proxy is total variance $\mathrm{RV}_t + r^2_{\text{overnight},t}$, written V_t in Seminar 9
  - A loss is **robust** if the ranking by *expected* loss with a proxy that is conditionally unbiased (given the forecasts' information) equals the ranking with true variance Patton (2011)
- Two robust losses
  - MSE $= (\mathrm{RV}_t - F_t)^2$: dominated by the most volatile days
  - QLIKE $= \dfrac{\mathrm{RV}_t}{F_t} - \ln \dfrac{\mathrm{RV}_t}{F_t} - 1 \geq 0$: depends only on the ratio; penalises under-prediction more
- Common mistake: MAE (mean absolute error), or MSE on volatility instead of variance, is **not** robust
  - Andersen & Bollerslev (1998): against r_t^2, GARCH forecasts explained a few % of the variation; against RV the same forecasts explain far more: the proxy was noisy, not the model

Comparing Two Forecasts: Diebold–Mariano

- Diebold & Mariano (1995) (DM) test: loss difference $d_t = L(\mathrm{RV}_t, F_t^A) - L(\mathrm{RV}_t, F_t^B)$; $t = \bar{d}/\mathrm{se}_{\mathrm{NW}}(\bar{d}) \to N(0,1)$
  - $L(\mathrm{RV}_t, F_t)$: the loss (QLIKE or MSE) of forecast F_t; $\bar{d}$: mean of d_t; $\mathrm{se}_{\mathrm{NW}}$: Newey–West standard error
  - $t < 0$: model A has the lower loss; HAC standard error because d_t is autocorrelated (Chapter 8)
- What drives t: the mean gap **and** how stable the gap is from day to day
  - Check, ignoring autocorrelation, $T = 100$ days: a gap with mean 0.01 and standard deviation 0.02 gives $t = 0.01/(0.02/\sqrt{100}) = 5$; a gap with mean 0.03 and standard deviation 0.30 gives $t = 1$
  - two forecasts that move together have a stable gap: a small mean difference can then be highly significant

Calibration: Mincer–Zarnowitz

- Mincer & Zarnowitz (1969) (MZ): $\mathrm{RV}_t = a + b\,F_t + u_t$: regress the realised target on the forecast (a: intercept, b: slope, u_t: error); an unbiased forecast has $a = 0$, $b = 1$; $b < 1$: forecasts too extreme
  - $b > 1$: forecasts too timid: realised variance moves more across days than the forecast does
  - Check: a forecast of 2 on calm days and 4 on busy days, with realised means 1.5 and 4.5, has $b = (4.5 - 1.5)/(4 - 2) = 1.5 > 1$ and $a = 1.5 - 1.5 \times 2 = -1.5$: it under-separates the two kinds of day
  - Joint Wald test with Newey–West covariance; R^2 measures how much of the variation the forecast explains
  - R^2 is capped by the noise in RV: even the true variance would not reach 1 Andersen & Bollerslev (1998)

Results: SPY, 2022–2026

Model	QLIKE	$\mathrm{DM}_{\mathrm{QLIKE}}$ t (p)	MSE (%^{4})	$\mathrm{DM}_{\mathrm{MSE}}$ p	MZ b	MZ R^2
log-HAR	0.310	–	4.15	–	1.16	0.30
HAR	0.326	-1.95 (0.052)	4.48	0.117	0.75	0.26
GARCH(1,1)-t	0.343	-2.30 (0.021)	4.87	0.188	0.70	0.20
EWMA	0.393	-4.02 (< 0.001)	5.14	0.050	0.74	0.14
Yesterday's RV	0.502	-5.95 (< 0.001)	5.41	0.060	0.54	0.29

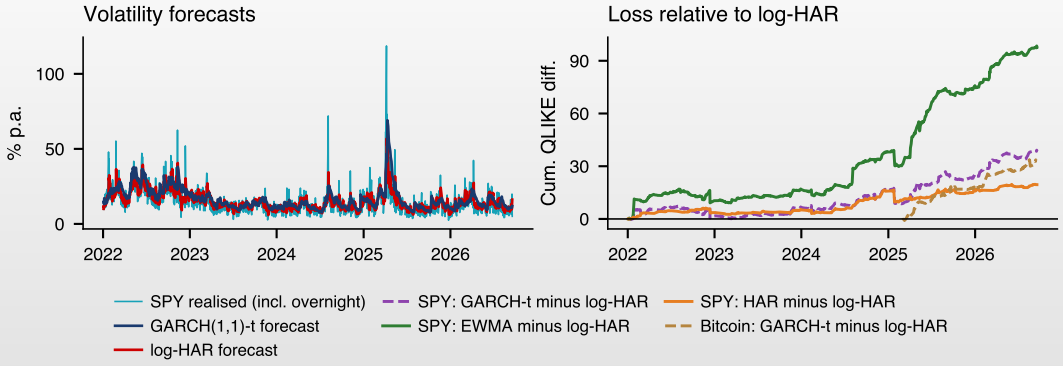
- 1 180 one-day-ahead forecasts of total daily variance; DM against log-HAR: $d_t = L(\text{log-HAR}) - L(\text{model})$ with $L = \text{QLIKE}$ or MSE, Newey–West; $t < 0$ favours log-HAR
- log-HAR has the lowest QLIKE and MSE: QLIKE 10% below GARCH(1,1)-t (DM $t = -2.30$, $p = 0.021$)
  - Against EWMA and yesterday's RV the gain is highly significant; against HAR in levels only marginally ($p = 0.052$)
- MSE differences are not significant at 5% (DM on MSE: p from 0.050 for EWMA to 0.188 for GARCH): a few crisis days (April 2025) dominate the squared errors
- Mincer–Zarnowitz: log-HAR $b = 1.16$ ($p = 0.554$ for $a = 0, b = 1$), GARCH $b = 0.70$: no significant bias, but log-HAR explains more (R^2 0.30 vs 0.20)
  - log-HAR $b > 1$: if anything slightly too timid, but only 0.7 standard errors (computed from unrounded estimates; SE 0.25) from 1; GARCH $b < 1$: slightly too extreme, also not significant
- Yesterday's RV has a high R^2 but $b = 0.54$: right direction, far too extreme

Results: Bitcoin, 2025–2026

Model	QLIKE	$\mathrm{DM}_{\mathrm{QLIKE}}$ t (p)	MSE (%^{4})	$\mathrm{DM}_{\mathrm{MSE}}$ p	MZ b	MZ R^2
log-HAR	0.282	–	26.6	–	0.82	0.38
HAR	0.409	-1.22 (0.223)	27.6	0.431	0.89	0.34
GARCH(1,1)-t	0.360	-4.32 (< 0.001)	42.2	0.055	0.55	0.23
EWMA	0.346	-2.96 (0.003)	36.3	0.033	0.64	0.18
Yesterday's RV	0.466	-5.38 (< 0.001)	35.0	0.061	0.58	0.33

- 422 one-day-ahead forecasts of 24-hour realised variance; same models and tests as for SPY
- log-HAR wins again: QLIKE 22% below GARCH(1,1)-t (DM $t = -4.32$, $p < 0.001$)
  - GARCH on daily data is biased: MZ $b = 0.55$ ($p < 0.001$): it overreacts to large daily moves
- HAR in levels is worse than GARCH under QLIKE (0.409 vs 0.360): after calm weekends it predicts too low a variance, and QLIKE punishes under-prediction
- Under MSE only EWMA is significantly worse than log-HAR at 5% ($p = 0.033$; GARCH $p = 0.055$)
- MSE is in $(\%^2)^2 = \%^4$ and grows with the square of the variance level, so Bitcoin and SPY MSE are not comparable; QLIKE depends only on RV_t/F_t and is comparable across markets

Forecasts over Time (1/2)



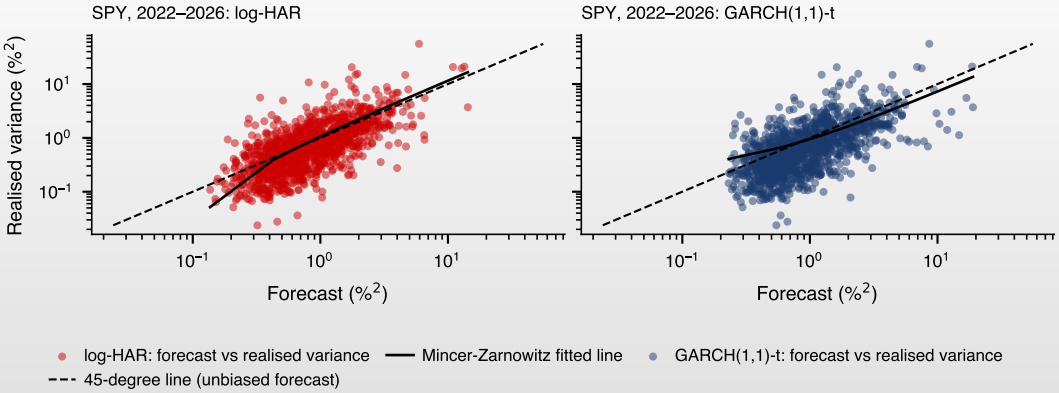
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Forecasts over Time (2/2)

- Left: SPY realised volatility (with overnight) and the two forecasts, annualised
  - right: cumulative QLIKE loss of each model minus log-HAR (rising = log-HAR better)
- After a shock both forecasts jump; GARCH stays high longer (its persistence is close to 1), log-HAR returns faster
  - April 2025: GARCH above 40% p.a. until late April
  - realised volatility below 20% from mid-April
- The cumulative loss difference rises steadily, with jumps at turbulent episodes: the gain is not driven by one day
- Of the three models plotted against log-HAR, EWMA loses most: a fixed $\lambda = 0.94$ reacts too much to single days

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Mincer–Zarnowitz Regressions (1/2)



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Mincer–Zarnowitz Regressions (2/2)

- SPY, 2022–2026: realised variance against the forecast (log scales); dashed: 45-degree line; black: fitted regression in levels
- log-HAR forecasts span a wider range: they separate calm from turbulent days better
- Only 1% of GARCH forecasts are below $0.26\%^2$ per day (log-HAR: minimum $0.14\%^2$): on calm days realised variance falls far below GARCH
- The cloud is wide for both: one day's variance is hard to predict, and RV itself is measured with error
- Weekly horizon and model extensions: Seminar 9 (B6, B7)

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Which Models Survive? The Model Confidence Set

Model	QLIKE	MCS p-value	Eliminated at step	In MCS (90%)
log-HAR	0.310	1.000	–	✓
SHAR	0.320	0.224	–	✓
HAR	0.326	0.014	3	
HARQ	0.344	0.095	4–5	
GARCH(1,1)-t	0.343	0.095	4–5	
EWMA	0.393	0.012	2	
Yesterday's RV	0.502	0.000	1	

- Model confidence set (MCS) Hansen, Lunde & Nason (2011): test equal predictive ability, drop the worst model, repeat (Seminar 9, B8)
  - range statistic, block bootstrap
  - the surviving set contains the best model with asymptotic probability $\geq 90\%$
- SPY, 1 180 days from 2022: the 90% set is {log-HAR, SHAR}
  - SHAR splits yesterday's RV into positive and negative realised semivariances Patton & Sheppard (2015)(▸ Appendix)
- A multiple-comparison procedure: many pairwise DM tests without size control would overstate the evidence (Chapter 8)

Reading the Model Confidence Set

- An MCS p-value is a running maximum: if a model is eliminated at step k, its MCS p-value is $\max_{j \le k} p_j$, with p_j the p-value of the equal-ability test at step j
  - so the MCS p-values rise along the elimination order, which the table reads off; equal values (GARCH, HARQ) mean the later model's own step p-value was not larger, so their relative order is not identified from the table
  - a model stays in the 90% set if its MCS p-value exceeds 0.10
- Why is HAR (QLIKE 0.326) dropped before GARCH (0.343)? Each step eliminates the model with the largest **standardised** loss gap to the other survivors, not the largest mean loss
  - SHAR differs from HAR only in its daily term, so their losses move together and the gap is very stable: DM SHAR vs HAR $t = -3.37$, $p < 0.001$
  - GARCH loses more on average, but its gaps are noisier: against log-HAR $t = -2.30$, smaller in size than the gap that separates HAR from SHAR (as in the check on the Diebold–Mariano slide)
  - the DM table against log-HAR cannot show this: it never pairs HAR with SHAR, whose very precise gap is the likely reason HAR goes before GARCH

Is Volatility Rough?

- Gatheral, Jaisson & Rosenbaum (2018): log volatility behaves like a **fractional Brownian motion** with Hurst exponent $H \approx 0.1$
  - $H = 0.5$: ordinary Brownian motion; $H < 0.5$: increments are negatively correlated, paths are rougher; $H > 0.5$: smoother, trending
- Scaling law: $m(q, \Delta) = \frac{1}{T} \sum_t |\ln \sigma_{t+\Delta} - \ln \sigma_t|^q \propto \Delta^{qH}$
  - $m(q, \Delta)$: the average q-th power of the change in log volatility over Δ days; σ_t: daily volatility (here $\sqrt{\mathrm{RV}_t}$); $\propto$: proportional to
  - Regress $\ln m(q, \Delta)$ on $\ln \Delta$: slope $\zeta_q = qH$; then H from the slopes across q
- Rough models reproduce the shape of the implied volatility smile for short maturities (Chapter 12)

Fractional Brownian Motion and the Scaling Law

- Fractional Brownian motion B_H, $0 < H < 1$: a Gaussian process with $B_H(0) = 0$, mean 0 and $\mathrm{Cov}(B_H(t), B_H(s)) = \frac{1}{2}\left(|t|^{2H} + |s|^{2H} - |t-s|^{2H}\right)$
  - $H = 1/2$: the covariance is $\min(t, s)$, ordinary Brownian motion
  - increments: $B_H(t+\Delta) - B_H(t) \sim N(0, \Delta^{2H})$ for every t
- Model $\ln \sigma_t = \text{const} + \nu B_H(t)$: the change over Δ days is $\nu \Delta^H Z$ in distribution, $Z \sim N(0,1)$
  - $\mathbb{E}|\ln \sigma_{t+\Delta} - \ln \sigma_t|^q = \nu^q \, \mathbb{E}|Z|^q \, \Delta^{qH}$, so $\ln m(q, \Delta) \approx \ln(\nu^q \mathbb{E}|Z|^q) + qH \ln \Delta$: the log–log slope is $\zeta_q = qH$
- Estimate: slopes ζ_q for $q \in \{0.5, 1, 1.5, 2, 3\}$, then regress ζ_q on q through the origin: $\hat{H} = \sum_q q\,\zeta_q / \sum_q q^2$
- Check with $q = 2$: for Brownian motion $m(2, \Delta) = \nu^2 \Delta$, so doubling Δ doubles m and $\zeta_2 = 1 = 2 \times 0.5$; with $H = 0.1$, doubling Δ multiplies $m(2, \Delta)$ only by $2^{0.2} \approx 1.15$

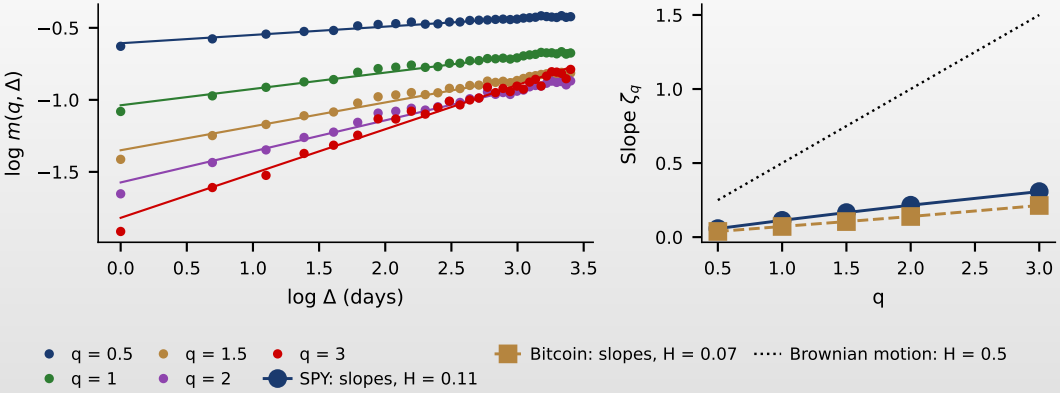
Rough or Noisy? The Identification Problem

- The estimate $H \approx 0.1$ comes from RV, a noisy proxy of the latent σ_t
  - Cont & Das (2024): the scaling estimator applied to realised volatility finds roughness even when spot volatility is smooth: an estimation artefact
  - Bolko, Christensen, Pakkanen & Veliyev (2023) (generalised method of moments, GMM, that models the RV error) and Fukasawa, Takabatake & Westphal (2022) (consistent under high-frequency asymptotics) still find $H < 0.5$
- Formal question: $\ln \mathrm{RV}_t^{1/2} = \ln \sigma_t + e_t$, with $\ln \sigma_t$ fractional of index H and e_t the measurement error of RV
  - H_0: $H = 1/2$ (classical diffusive volatility) against H_1: $H < 1/2$
  - Identification problem: a small H and a large $\mathrm{Var}(e_t)$ both flatten $m(2, \Delta)$ at short lags, because $m(2, \Delta) = c\,\Delta^{2H} + 2\,\mathrm{Var}(e_t)$ for i.i.d. error (next slide)
- The scaling estimator on noisy RV proxies is **inconsistent** for the Hurst exponent; consistency needs high-frequency asymptotics Fukasawa, Takabatake & Westphal (2022)
  - Short-run roughness can coexist with long memory Bennedsen, Lunde & Pakkanen (2022)
- Stakes: the short-maturity skew and volatility forecasts depend on H (Chapter 12)

How Measurement Error Flattens the Scaling Law

- Observed increment = latent increment + error increment: $(\ln \sigma_{t+\Delta} - \ln \sigma_t) + (e_{t+\Delta} - e_t)$
  - if e_t is i.i.d. and independent of σ, the variances add: $\mathrm{Var}(e_{t+\Delta} - e_t) = 2\,\mathrm{Var}(e_t)$ for every $\Delta \geq 1$
  - hence $m(2, \Delta) = c\,\Delta^{2H} + k$, $c = \nu^2$, $k = 2\,\mathrm{Var}(e_t)$: a constant that does not grow with Δ
- Toy: $H = 0.5$, $c = 1$, $k = 1$
  - $m(2, 1) = 1 + 1 = 2$, $m(2, 2) = 2 + 1 = 3$: log–log slope $\ln(3/2)/\ln 2 = 0.58$, so $\hat{H} = 0.58/2 = 0.29$ instead of 0.5
  - at long lags the constant matters less: from $\Delta = 20$ to 40, slope $\ln(41/21)/\ln 2 = 0.97$, $\hat{H} = 0.48$
- Testable implication: under i.i.d. error, $\hat{H}$ rises when short lags are dropped; the mini-case in section AI for Scientific Discovery runs exactly this check on SPY

Scaling of Log Volatility Increments (1/2)



MFM_ch9_rough

Scaling of Log Volatility Increments (2/2)

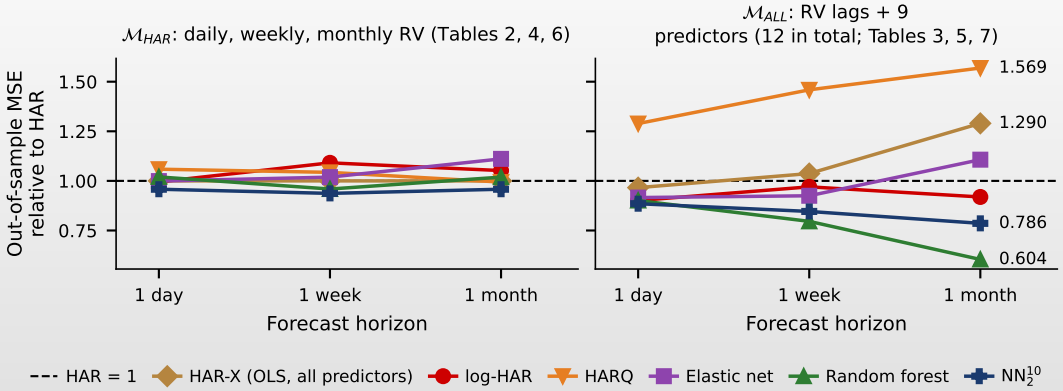
- Left: $\ln m(q, \Delta)$ against $\ln \Delta$ for SPY, $\Delta = 1, \ldots, 30$ days; right: slopes ζ_q against q
- The log–log points lie close to straight lines and the slopes are close to linear in q: consistent with the scaling law, which the plot alone cannot establish
  - $\hat{H} = 0.11$ for SPY (block bootstrap 95% CI $[0.08; 0.13]$), 0.07 for Bitcoin (only observed pairs at each calendar lag; interpolating the missing days gives 0.08): far below 0.5
- Caution: RV measures volatility with error; i.i.d. error lowers $\hat{H}$, other error structures can bias it either way, so $\hat{H}$ is not a bound on the latent H
  - A simulated Brownian motion gives $\hat{H} = 0.46$; plus noise of a similar relative size, $\hat{H} = 0.18$ (Seminar 9, B9)

MFM_ch9_rough

Case Study: Can Machine Learning Beat HAR?

- Christensen, Siggaard & Veliyev (2023): research questions
  - does off-the-shelf machine learning (ML) forecast daily realised variance better than the HAR family?
  - if so, why?
  - HAR Corsi (2009) is linear in three RV averages, while RV is right-skewed and persistent: a nonlinear map, or predictors beyond past RV, may help
- Data: 29 of the 30 stocks of the Dow Jones Industrial Average (DJIA), January 2001 – December 2017, $T = 4\,257$ days, 5-minute RV ($n = 78$)
  - Chronological split: training 70%, validation 10%, test 20%
- Information sets: $\mathcal{M}_{\mathsf{HAR}}$ (daily, weekly, monthly RV) and $\mathcal{M}_{\mathsf{ALL}}$, which adds 9 predictors: implied volatility, VIX, volume, momentum, macro and uncertainty indices (Table 1)
- Models, with minimal tuning: the HAR family by OLS, penalised regressions, tree ensembles (bagging, random forest, boosting) and neural networks; full catalogue in the ▸ Appendix
- Evaluation: out-of-sample MSE relative to HAR, per stock and then averaged; one-sided DM test per stock; MCS at 90%

Case Study: Gains by Horizon (1/2)



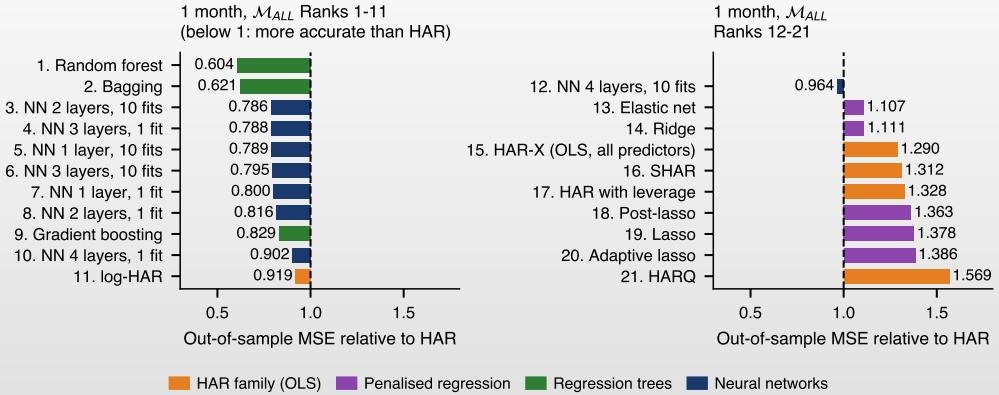
MFM_ch9_ml_vs_har

Case Study: Gains by Horizon (2/2)

- Source: Christensen, Siggaard & Veliyev (2023), Tables 2–7: out-of-sample MSE relative to HAR, averaged over 29 DJIA stocks (test set: the last 20% of 2001–2017); left: RV lags only, right: all 12 predictors; HAR-X: OLS on all predictors
- RV lags only: the six models shown stay between 0.937 and 1.111 times the HAR MSE at all three horizons: three RV averages leave little for a flexible model to learn
- All 12 predictors: the gain of trees and networks grows with the horizon
  - random forest 0.901 $\rightarrow$ 0.796 $\rightarrow$ 0.604
  - network with 2 hidden layers, averaged over 10 fits (NN_2^{10}): 0.885 $\rightarrow$ 0.846 $\rightarrow$ 0.786
  - the authors attribute it to the higher persistence of the ML fits (Figure 8)
- Linear models with the same predictors deteriorate (HAR-X 0.966 $\rightarrow$ 1.290, HARQ 1.289 $\rightarrow$ 1.569); only log-HAR stays below 1 (0.901, 0.970, 0.919)

MFM_ch9_ml_vs_har

Case Study: Ranking of 21 Models, One Month Ahead (1/2)



MFM_ch9_ml_vs_har

Case Study: Ranking of 21 Models, One Month Ahead (2/2)

- Source: Christensen, Siggaard & Veliyev (2023), Table 7
  - MSE relative to HAR, one month ahead, all 12 predictors, mean over 29 DJIA stocks
  - NN_k^e: k hidden layers, e networks averaged
- Tree ensembles lead: random forest 0.604, bagging 0.621
  - all eight networks beat HAR (0.786–0.964), but depth does not help
- Penalised regressions and linear HAR extensions lose to HAR
  - 1.107–1.569; the one linear exception, log-HAR, is on the horizon chart
- Each bar averages per-stock MSE ratios
  - not a significance test of the ranking; on SPY and under QLIKE: Seminar 9, C3

MFM_ch9_ml_vs_har

The Discovery Loop on the Roughness Question

- Question: is $\hat{H} \approx 0.1$ a property of σ_t or of the RV error e_t? H_0: $H = 1/2$ against H_1: $H < 1/2$
- One pass of the loop with an AI assistant (reusable prompts in the Appendix)
  - Literature: tabulate the published estimates of H (estimator, frequency, treatment of the RV error); check every DOI (digital object identifier) on Crossref
  - Competing hypotheses, each with its shape of $m(2, \Delta)$: fractional with $H = 0.1$; $H = 0.5$ plus i.i.d. error; two factors, one fast mean-reverting
  - Replicate the chapter's $\hat{H} = 0.11$ before extending; fix lag windows, measures and the fit with a constant noise term k in advance
- Checks the researcher owns
  - Proxy error: simulate $H = 0.5$ plus error of the same size; the procedure must not return $H \approx 0.1$
  - Overlapping increments invalidate OLS standard errors of ζ_q: use a block bootstrap; report every window tried
  - Assistants quote "$H \approx 0.1$ for all assets": check each number against the cited table

Mini-Case: Is $\hat{H}$ an Artefact of i.i.d. Error?

- A typical assistant proposal: $\hat{H} \approx 0.1$ comes from i.i.d. measurement error in RV. Implication: $m(2, \Delta) = c\,\Delta^{2\hat{H}} + k$ with $k = 2\,\mathrm{Var}(e_t) > 0$, so $\hat{H}$ **rises** when short lags are dropped

	$\Delta \in [1, 30]$	$[5, 30]$	$[10, 50]$	Fit with k, $[1, 50]$
Simulated $H = 0.5$ plus error	0.18	0.24	0.26	0.39
SPY, total RV	0.106	0.089	0.077	0.096
SPY, intraday RV	0.129	0.094	0.078	0.113

- First column replicates the chapter; the simulated row (one path, not a power study) shows the mechanism: $\hat{H}$ rises and the fit recovers most of the gap to 0.5
- SPY: $\hat{H}$ **falls** with the window and the fitted k is 0; $\hat{H} = 0.096$, block bootstrap 95% CI $[0.058; 0.268]$
- Interpretation: an illustrative diagnostic, inconsistent with the i.i.d.-error explanation; a formal rejection needs a simulation-calibrated test; the non-i.i.d. mechanism of Cont & Das (2024) is not tested by this design.

MFM_ch9_ai_discovery_case

Your Turn: A Project Seed

- Question: does the roughness of SPY and Bitcoin volatility survive an estimator that models the RV measurement error?
- Replicate first: $\hat{H} = 0.11$ (SPY) and 0.07 (Bitcoin) from the scaling law, and $\hat{H} = 0.18$ for simulated $H = 0.5$ plus error (Seminar 9, B9)
- Deliverables
  - A pre-registration (hypotheses, estimators, lag windows, tests, simulation design) dated before the first run
  - The GMM estimator of Bolko, Christensen, Pakkanen & Veliyev (2023) on the course RV series, with Monte Carlo size and power at $H \in \{0.1, 0.5\}$
  - Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Log every prompt that shaped a design choice in `AI_USE.md` and every error the assistant made in `AI_ERRORS.md`: github.com/danpele/MFM/tree/main/project

Which Tool When?

Situation	Suggested tool
Liquid asset, 1–5 minute bars	RV at 5 minutes, subsampled
Tick data, or an illiquid stock (Bucharest Stock Exchange, BVB)	Pre-averaging or flat-top realised kernel ($n^{-1/4}$; Parzen $n^{-1/5}$; Appendix: rates of convergence); check the signature plot
Daily risk of an exchange-traded asset	Total variance: RV plus squared overnight return
Separate news jumps from diffusion	BV or MedRV (median realised variance; Appendix: jump-robust estimators), ratio test with FDR control; Lee–Mykland for the jump time
Daily or weekly variance forecast	log-HAR (plus jumps, weekends, RQ); benchmark GARCH
Comparing forecasts	QLIKE and MSE on variance, Diebold–Mariano, Mincer–Zarnowitz, model confidence set

- Strongest result: out of sample, log-HAR beats GARCH(1,1)-t under QLIKE by 10% (SPY) and 22% (Bitcoin); MSE comparisons are inconclusive
- Assumptions: the 5-minute rule rests on a flat signature plot at 5 minutes and coarser; the jump counts on FDR control and the periodicity correction

Check Yourself, and Next: Chapter 10

Retrieval questions

- Why is R^2 of log RV not comparable with R^2 of RV in levels? (*Reading the HAR Estimates*)
- Why is QLIKE robust to a noisy RV proxy while MSE on volatility is not? (*Evaluating Against a Noisy Target*)
- Why does the MCS state a set of models rather than a winner? (*Which Models Survive?*)
- How can measurement error in RV make volatility look rough? (*How Measurement Error Flattens the Scaling Law*)

Chapter 10: market microstructure

- Where does the noise ε of this chapter come from: limit order books, bid–ask spreads, tick sizes
- Liquidity, market impact and high-frequency trading, on equity and crypto exchanges

Itô Semimartingales and Stable Convergence

- Itô semimartingale: $dp = \mu\, ds + \sigma\, dW + dJ$, with μ, σ adapted (known from the past at each time) and σ càdlàg (right-continuous with left limits, so it may jump); σ may depend on W (leverage)
  - Without jumps: $\sqrt{M}(\mathrm{RV}_t - \mathrm{IV}_t) \xrightarrow{\mathcal{L}\text{-}s} \mathrm{MN}(0, 2\,\mathrm{IQ}_t)$, a **mixed** Normal distribution whose variance $2\,\mathrm{IQ}_t$, with integrated quarticity IQ, $\mathrm{IQ}_t = \int \sigma^4(s)\, ds$, is itself random Barndorff-Nielsen & Shephard (2002)
- **Stable** convergence ($\mathcal{L}$-s): the limit holds jointly with every variable of the path's σ-field Mykland & Zhang (2009)
  - Hence $(\mathrm{RV}_t - \mathrm{IV}_t)/\sqrt{2\,\mathrm{RQ}_t/M} \to N(0,1)$: studentising by $\sqrt{2\,\mathrm{RQ}_t/M}$, with $\mathrm{RQ}_t \to \mathrm{IQ}_t$ random, is legitimate only under stable convergence
- Finite samples: the t-statistic for $\ln \mathrm{RV}_t$ has better coverage than the one in levels Barndorff-Nielsen & Shephard (2005); the confidence intervals of the lecture use it

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Deriving the Noise Bias and the Optimal Frequency

- Observed return $r_i = r_i^* + \varepsilon_i - \varepsilon_{i-1}$, with ε i.i.d. $(0, \omega^2)$ and independent of p^*
  - Given the path of σ (zero drift, no leverage: σ independent of W):
    $\mathbb{E}[\mathrm{RV}^{(n)} \mid \sigma] = \sum_i \mathbb{E}[r_i^{*2} \mid \sigma] + \sum_i \mathbb{E}[(\varepsilon_i - \varepsilon_{i-1})^2] = \mathrm{IV} + 2n\omega^2$; $\mathrm{cov}(r_i, r_{i-1}) = -\mathbb{E}[\varepsilon_{i-1}^2] = -\omega^2$
  - By a law of large numbers for the noise terms, $\mathrm{RV}^{(n)}/(2n) \xrightarrow{p} \omega^2$ as $n \to \infty$: the all-tick RV estimates the noise, not IV Bandi & Russell (2008)
- MSE: $\mathbb{E}[(\mathrm{RV}^{(n)} - \mathrm{IV})^2] \approx \underbrace{2\,\mathrm{IQ}/n}_{\text{discretisation}} + \underbrace{4n^2\omega^4}_{\text{squared bias}} + O(n)$
  - First-order condition $-2\,\mathrm{IQ}/n^2 + 8n\omega^4 = 0$ gives $n^* = (\mathrm{IQ}/(4\omega^4))^{1/3}$ Bandi & Russell (2008)

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Intraday Seasonality and the Jump Test

- Seasonality biases BV (not RV): $\mathbb{E}|r_i|\,|r_{i-\mathbf{1}}| \propto \sigma_i \sigma_{i-\mathbf{1}} \leq (\sigma_i^{\mathbf{2}} + \sigma_{i-\mathbf{1}}^{\mathbf{2}})/2$
  - Fast changes of scale (open, close) push $\mathrm{RV} - \mathrm{BV}$ up, so the ratio test (section Jumps) over-rejects
- Periodicity factor f_i, standard-deviation estimator of Boudt, Croux & Laurent (2011): scale each return by $\sqrt{\mathrm{BV}_t / M_t}$, take the root mean square by interval of the day, normalise the mean of $f_i^{\mathbf{2}}$ to 1
  - SPY: $f = 1.78$ at the open, 0.73 at the minimum, 1.52 at the close
- Dividing $r_{t,i}$ by f_i before the test (5% level: main text): at the 0.1% level 20 SPY rejections remain

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Rates of Convergence under Noise

Estimator	Error of order	Positive	Source
RV on all ticks	inconsistent (bias $2n\omega^2$)	yes	Bandi & Russell (2008)
Sparse RV, fixed 5-minute grid	$M^{-1/2}$, M fixed; bias stays	yes	Liu, Patton & Sheppard (2015)
Two-scale RV	$n^{-1/6}$	no	Zhang, Mykland & Aït-Sahalia (2005)
Multi-scale RV	$n^{-1/4}$	no	Zhang (2006)
Realised kernel, flat-top / Parzen	$n^{-1/4}$ / $n^{-1/5}$	no / yes	Barndorff-Nielsen, Hansen, Lunde & Shephard (2008)
Pre-averaging	$n^{-1/4}$	no	Jacod, Li, Mykland, Podolskij & Vetter (2009)

- $n^{-1/4}$ is the optimal rate with i.i.d. noise: the price of noise is half the usual exponent
- With $n = 23\,400$ seconds: $n^{1/4} \approx 12.4$, against $\sqrt{78} \approx 8.8$ for 5-minute RV: a heuristic only (rates ignore constants, noise size and the bias of sparse RV), consistent with the modest gains in Liu, Patton & Sheppard (2015)

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Pre-Averaging

- Local averages of $k_n \approx \theta\sqrt{n}$ returns (θ: a tuning constant, not the jump-test θ) with a weight g on $[0,1]$, e.g. $g(x) = \min(x, 1-x)$ Jacod, Li, Mykland, Podolskij & Vetter (2009)
  - $\bar{r}_i = \sum_{j=1}^{k_n - 1} g(j/k_n)\, r_{i+j}$: noise variance shrinks like $1/k_n$, the signal keeps order k_n/n
- Estimator, with $\psi_1 = \int_0^1 g'(x)^2 dx$, $\psi_2 = \int_0^1 g(x)^2 dx$ (1 and 1/12 for the triangular g; g': the derivative of g)
  - $\widehat{\mathrm{IV}} = \frac{1}{\theta\, \psi_2 \sqrt{n}} \sum_i \bar{r}_i^2 - \frac{\psi_1}{2\,\theta^2 \psi_2\, n} \sum_i r_i^2$ (second term: noise bias correction)
- Rate $n^{-1/4}$; the same device gives jump-robust (pre-averaged bipower) and covariance estimators
- Built for tick data: on 5-minute bars little noise is left to average out (empirical signature plot, section Microstructure Noise)

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Realised Covariance and Asynchronous Trading

- Realised covariance $\sum_i r^{(1)}_{t,i} r^{(2)}_{t,i} \to \int \sigma_{12}(s)\, ds$ needs prices of both assets on the **same** grid; $r^{(1)}, r^{(2)}$: the returns of assets 1 and 2, $\sigma_{12}(s)$: their spot covariance
  - Epps effect Epps (1979): measured correlation falls towards 0 as sampling gets finer, because trades are asynchronous and the last price is stale
- Hayashi & Yoshida (2005): sum the products of **all** pairs of returns whose time intervals overlap
  - $\sum_{i,j} r^{(1)}_i r^{(2)}_j \, 1\{(t_{i-1}, t_i] \cap (s_{j-1}, s_j] \neq \emptyset\}$: $(t_{i-1}, t_i]$, $(s_{j-1}, s_j]$: the intervals between consecutive trades of each asset; unbiased without synchronisation (in the absence of noise)
- With noise as well: pre-averaged or multivariate kernel versions; application: intraday portfolio risk (Chapter 4)

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Locating Jumps Within the Day, Lee–Mykland

- Lee & Mykland (2008): $L_{t,i} = r_{t,i}/\hat{\sigma}_{t,i}$, with $\hat{\sigma}^2_{t,i}$ the bipower average of the previous $K = 270$ returns (their value for 5-minute data)
  - Under no jumps, $\max_i |L_i|$ over n returns has a Gumbel limit: reject if $(|L| - C_n)/S_n > -\ln(-\ln(1-\alpha))$, with $C_n = \sqrt{2\ln n} - \frac{\ln\pi + \ln\ln n}{2\sqrt{2\ln n}}$ and $S_n = 1/\sqrt{2\ln n}$, the centring and scaling constants of the Gumbel limit (our $\hat{\sigma}^2$ includes the factor $\pi/2$, so the constant $c = \sqrt{2/\pi}$ of Lee & Mykland (2008) becomes 1)
  - SPY, returns divided by f_i, $\alpha = 1\%$, $n = 115\,453$ (the whole sample): threshold $|L| > 5.41$
- 169 jumps on 105 days; 62 of them on Wednesdays between 14:00 and 15:00, when Federal Open Market Committee (FOMC) statements (14:00) and press conferences (14:30) are released
  - Largest: $L = -22.9$ on 2025-10-29; only 10% of these days are also flagged by the daily ratio test
- The two tests ask different questions: one extreme return relative to local volatility, versus a daily gap between RV and BV

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Jump-Robust Estimators of Integrated Variance

- Truncated RV Mancini (2009): $\mathrm{TRV}_t = \sum_i r_{t,i}^{\mathbf{2}} \, 1\{|r_{t,i}| \le c\,\hat{\sigma}_t f_i\,\Delta^{\varpi}\}$, $\varpi \in (0, 1/2)$
  - Keep only the squared returns below a threshold: c: a multiple, $\hat{\sigma}_t$: a jump-robust daily volatility, f_i: the periodicity factor, ϖ: the exponent of the interval length Δ
  - Diffusive increments are $O_p(\Delta^{\mathbf{1/2}})$ (of order $\sqrt{\Delta}$) and end up below the threshold; jumps are $O(1)$ and are cut off
- MedRV Andersen, Dobrev & Schaumburg (2012): $\frac{\pi}{\mathbf{6-4\sqrt{3}+\pi}} \frac{M}{M-\mathbf{2}} \sum_i \mathrm{med}(|r_{i-\mathbf{1}}|, |r_i|, |r_{i+\mathbf{1}}|)^{\mathbf{2}}$
  - med: the median of three adjacent absolute returns; the first fraction makes MedRV consistent for IV
  - An isolated return larger than both neighbours never enters the median (which shifts to a neighbour), so a large isolated jump is excluded, though its effect is not removed exactly in finite samples; MedRV is less sensitive than BV to isolated zero returns; two adjacent jumps can still enter the median
- Jump share of SPY RV: BV 6.3%, MedRV 7.2%, TRV 12.5% ($c = 3$, $\varpi = 0.49$; truncation bites on 44% of days: c is a tuning choice)
- Aït-Sahalia & Jacod (2009): compare power variations $\sum |r|^p$, $p > 2$, at Δ and $k\Delta$; the ratio tends to 1 with jumps and to $k^{p/\mathbf{2}-\mathbf{1}}$ without: known limits under both hypotheses

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HAR as a Restricted AR(22), the Wald Test

- Unrestricted: $\ln \mathrm{RV}_{t+1} = \phi_0 + \sum_{k=1}^{22} \phi_k \ln \mathrm{RV}_{t+1-k} + \varepsilon_{t+1}$
  - HAR imposes $\phi_2 = \ldots = \phi_5$ (3 restrictions) and $\phi_6 = \ldots = \phi_{22}$ (16): $R\phi = 0$ with 19 rows
- Wald: $W = (R\hat{\phi})^\top (R\hat{V}_{\mathrm{NW}} R^\top)^{-1} R\hat{\phi} \to \chi^2_{19}$ under the HAR restrictions; R: the 19×23 restriction matrix, $\hat{V}_{\mathrm{NW}}$: Newey–West covariance of $\hat{\phi}$; large W = restrictions rejected
- SPY log total variance: $W = 18.1$, $p = 0.51$; R^2 0.477 (AR(22)) vs 0.470 (HAR)
  - The step restrictions are not rejected: HAR saves 19 parameters at almost no in-sample cost, and less estimation noise helps out of sample

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Measurement Error in the Regressors and HARQ

- $\mathrm{RV}_t = \mathrm{IV}_t + u_t$ with $\mathbb{E}[u_t \mid \sigma] = 0$ and $\mathrm{Var}(u_t \mid \sigma) = 2\,\mathrm{IQ}_t/M$ (the CLT of the section Realised Variance)
  - If $\mathrm{IV}_{t+\mathbf{1}} = \beta_{\mathbf{0}} + \beta\,\mathrm{IV}_t + \eta_{t+\mathbf{1}}$, OLS of $\mathrm{RV}_{t+\mathbf{1}}$ on RV_t gives plim $\hat{\beta} = \beta\,\dfrac{\mathrm{Var(IV)}}{\mathrm{Var(IV)} + \mathbb{E}[2\,\mathrm{IQ}/M]} < \beta$ for $\beta > 0$: attenuation ($\eta_{t+\mathbf{1}}$: the shock to IV; plim: probability limit)
- If the signal variance does not rise with IQ_t (a working approximation), the signal-to-noise ratio of RV_t falls when IQ_t is high; HARQ approximates the falling weight by $\beta_d + \beta_Q\sqrt{\mathrm{RQ}_t}$, $\beta_Q < 0$ Bollerslev, Patton & Quaedvlieg (2016)
- SPY, levels, Newey–West SE: intraday RV $\hat{\beta}_Q = -0.0088$ (0.0042); total variance -0.0022 (0.0024)
  - Full vector $(\hat{\beta}_{\mathbf{0}}, \hat{\beta}_d, \hat{\beta}_w, \hat{\beta}_m)$, SE in brackets; intraday RV: 0.015 (0.105), 1.135 (0.494), -0.120 (0.228), 0.026 (0.057); total variance: 0.217 (0.109), 0.542 (0.282), 0.173 (0.128), 0.088 (0.067)
  - Daily slope (intraday RV) 1.13 at the median $\sqrt{\mathrm{RQ}}$, 1.09 at its 99th percentile: the sign predicted by the theory, a small effect
  - Read $\hat{\beta}_d$ together with the other slopes: the daily slope above 1 is partly offset by a negative weekly slope, and the three RV averages overlap by construction (each contains yesterday's RV), so the split is imprecise (SE of $\hat{\beta}_d$: 0.494); the slopes sum to 1.04 (total variance: 0.80), so the intraday levels fit is an in-sample diagnostic for the sign of β_Q, not a forecasting model
  - Like for like, total variance: HAR $\hat{\beta}_d = 0.399$, HARQ $\hat{\beta}_d = 0.542$: letting the weight fall with RQ raises the base daily slope, in the direction the attenuation argument predicts (the difference is within one SE)

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Beyond HAR, Semivariances and Joint Return–RV Models

- Realised semivariances $\mathrm{RS}_t^{\pm} = \sum_i \boldsymbol{r}_{t,i}^{\boldsymbol{2}} 1\{r_{t,i} \gtrless 0\}$, $\mathrm{RS}^+ + \mathrm{RS}^- = \mathrm{RV}$; SHAR Patton & Sheppard (2015): $\beta_d \mathrm{RV}_t$ becomes $\beta^+ \mathrm{RS}_t^+ + \beta^- \mathrm{RS}_t^-$
  - SPY, levels (overnight return counted as one more return of the day): $\hat{\beta}^- = 0.56$ (0.23), $\hat{\beta}^+ = 0.21$ (0.18); Wald $\beta^+ = \beta^-$: $p = 0.35$
  - Bad volatility is more persistent, as in Patton & Sheppard (2015), but six years of data do not separate the two slopes
  - In the MCS comparison, HARQ and SHAR are estimated in levels with the filter of Bollerslev, Patton & Quaedvlieg (2016): forecasts outside the [min, max] range of the estimation-sample values are replaced by the sample mean
- Realized GARCH Hansen, Huang & Shek (2012): a GARCH equation driven by the realised measure x_t plus a **measurement equation**
  - $\ln h_t = \omega + \beta \ln h_{t-\boldsymbol{1}} + \gamma \ln x_{t-\boldsymbol{1}}$, $\ln x_t = \xi + \varphi \ln h_t + \tau(z_t) + u_t$, $r_t = \sqrt{h_t}\, z_t$: joint quasi-maximum likelihood
  - h_t: conditional variance; x_t: realised measure (e.g. RV); z_t: standardised shock; $\tau(z_t)$: leverage function; ξ, φ: measurement-equation coefficients, u_t: its error
- HEAVY (high-frequency-based volatility) models Shephard & Sheppard (2010): one GARCH-type equation for the daily variance, driven by lagged RV, and one for RV itself: consistent multi-step forecasts

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Case Study Data, 29 DJIA Stocks and 12 Predictors

- Data (paper, Data description): 29 of the 30 DJIA stocks (Visa excluded), 29 January 2001 – 31 December 2017, $T = 4\,257$ days; 5-minute RV from cleaned New York Stock Exchange (NYSE) trade prices, $n = 78$ returns a day
  - Split: training 70% (2 964 days), validation 10% (424), test 20% (847); robustness: training 1 000 or 2 000 days, validation 200 (paper, Appendix A.1)
  - $\mathcal{M}_{\mathbf{HAR}}$: daily, weekly, monthly RV; $\mathcal{M}_{\mathbf{ALL}}$ adds 9 predictors (Table 1): implied volatility, earnings-day dummy, 1-week momentum, change in log dollar volume; VIX, squared Hang Seng return, business-conditions index, change in the 3-month T-bill rate, policy-uncertainty index

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Case Study Models, from HAR to Neural Networks

- Models (paper, Section 1): HAR, log-HAR, HAR with leverage, SHAR, HARQ by OLS on a rolling window; ridge (L_2 penalty), lasso (L_1), elastic net (a mix), adaptive and post-lasso, penalties tuned on the validation set
  - Trees: bagging, random forest (RF) with Breiman's defaults, 500 trees, $J/3$ candidate predictors per split (J: number of predictors), minimum node size 5 (paper, Table A.6), gradient boosting
  - Neural networks NN_k^e: $k = 1, \ldots, 4$ hidden layers (2; 4-2; 8-4-2; 16-8-4-2 neurons), leaky ReLU, Adam, dropout, early stopping; $e = 1$: best of 100 seeds on validation, $e = 10$: average of the best 10
- Evaluation (paper, Section 1.6): out-of-sample MSE relative to HAR, averaged over stocks; one-sided DM test per stock Diebold & Mariano (1995); MCS at 90% Hansen, Lunde & Nason (2011)

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Case Study, Further Results

- One day ahead, RV lags only (Table 2): MSE relative to HAR: log-HAR 0.995, HARQ 1.059, RF 1.020, neural networks 0.954–0.990
  - Past RV leaves little to learn: the networks give the clearest gains (1–5%), log-HAR barely beats HAR
- One day ahead, all 12 predictors (Table 3): elastic net 0.916; the other models are on the horizon chart of the case study
  - Extra predictors pay most with shrinkage, nonlinearity or the log transform; HARQ overfits
- MCS at 90% (Figure 4): with all predictors, the networks stay in the set for about 80% of stocks, HAR for about 20–40%
  - Variable importance from accumulated local effects (Figure 7): daily RV, weekly RV and implied volatility dominate in all four models shown; RF and networks rank implied volatility higher than OLS with all predictors and elastic net

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Case Study, Replication, Limits and Extensions

- Replicate on course data: SPY 5-minute bars, open-to-close RV with 78 returns a day (the sampling in the Data description section); the authors report that SPY confirms the stock results (footnote 12)
  - Split from Appendix A.1 of the paper: 1 000 training days, 200 validation days, the rest to 18 September 2026 as test set
  - Predictors: VIX (computed from S&P 500 index options; a proxy for SPY's own implied volatility, which would need SPY options), squared Hang Seng return, SPY momentum and dollar volume, T-bill and policy uncertainty (Federal Reserve Economic Data, FRED); no earnings dummy for a fund (Seminar 9, C3)
- Limits stated by the authors
  - Minimal tuning: more tuning could improve ML further (paper, Section 6); the networks are trained once and not re-estimated in the test window (paper, Section 2)
  - Rankings depend on the split: with 1 000 training days the financial crisis falls in the test set and the HAR extensions beat HAR (paper, Section 3.1 and Appendix A.1)
  - HAR models with a separate jump component are not included (paper, Section 1.2); accumulated local effects describe the average prediction and miss rare events such as earnings days (paper, Section 3.2)
- Open extension: the paper ranks variance forecasts by MSE only
  - does the ML advantage survive QLIKE, which is robust to the noisy RV proxy Patton (2011)?
  - does it hold for Bitcoin's 24-hour RV?

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The Discovery Loop, Reusable Prompts (1/2)

- Tools: an AI assistant (a large language model, LLM, such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: Semantic Scholar, Elicit, Google Scholar
- **1. Literature**: tabulate the evidence, then check every DOI on Crossref (the title must match)
  - `Tabulate the estimates of H published since 2018: estimator, sampling frequency, treatment of the RV measurement error, point estimate and standard error, DOI.`
- **2. Competing hypotheses**, each with its implied shape of $m(2, \Delta)$
  - `Derive m(2, Delta) for three processes that give H near 0.1 with the scaling estimator: fractional log-volatility with H = 0.1; H = 0.5 plus i.i.d. error; a two-factor model with one fast mean-reverting factor.`
- **3. Data and code**: replicate the chapter's number first
  - `Write Python that builds daily total variance from the SPY 5-minute bars of the course data (5-minute RV plus the squared overnight return, previous close to today's open), computes m(q, Delta) for q in {0.5, 1, 1.5, 2, 3}, Delta = 1..30, and the slope H.` Target: $\hat{H} = 0.11$

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The Discovery Loop, Reusable Prompts (2/2)

- **4. Robustness**, pre-registered, with a power check
  - `Write a pre-registered plan: lag windows [1,30], [5,30], [10,50]; RV, bipower variation and 30-minute subsampled RV; a fit m(2, Delta) = c Delta^(2H) + k; block-bootstrap intervals; size and power by simulation at H = 0.1 and 0.5.`
- **5. Critique**: ask for the strongest counter-argument
  - `Act as a referee: which non-i.i.d. error structures in log RV (serially correlated, level-dependent, driven by overnight returns or jumps) would survive the constant-k correction, and how would you detect each?`
- **6. Report**: all windows and measures, bootstrap intervals, the simulation design, code and the AI log

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Further Reading

- Textbook and surveys
  - Franke, Härdle & Hafner (2019): statistics of financial markets, volatility models
  - Andersen, Bollerslev, Diebold & Labys (2003): modelling and forecasting realised volatility; Corsi (2009): the HAR model
- Crypto-assets
  - Intraday information can enter Bitcoin risk forecasts; the entropy of high-frequency returns has been proposed as one such signal Pele & Mazurencu-Marinescu-Pele (2019)
  - Crypto-assets behave like a distinct asset class with their own volatility dynamics Pele et al. (2023)

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Further Study on Quantinar

Advanced courses on quantinar.com

- Nonparametric and Semiparametric Models
- Kalman Filter
- Applied Time Series Analysis with Python

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_09

Acronyms used in this material (1/2)

- **ACF**: AutoCorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **BV**: Bipower Variation
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CI**: Confidence Interval
- **CLT**: Central Limit Theorem
- **DJIA**: Dow Jones Industrial Average
- **DM**: Diebold–Mariano (test)
- **ETF**: Exchange-Traded Fund
- **EWMA**: Exponentially Weighted Moving Average
- **FDR**: False Discovery Rate
- **FOMC**: Federal Open Market Committee
- **FRED**: Federal Reserve Economic Data
- **FWER**: Family-Wise Error Rate
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GMM**: Generalized Method of Moments
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HAR-CJ**: HAR model with Continuous and Jump components
- **HAR-RV**: Heterogeneous AutoRegressive model of Realised Volatility

Acronyms used in this material (2/2)

- **HARQ**: HAR model with Realised Quarticity
- **HEAVY**: High-frEquency-bAsed VolatilitY (models)
- **IQ**: Integrated Quarticity
- **LLM**: Large Language Model
- **MAE**: Mean Absolute Error
- **MCS**: Model Confidence Set
- **ML**: Machine Learning
- **MSE**: Mean Squared Error
- **MZ**: Mincer–Zarnowitz (regression)
- **NYSE**: New York Stock Exchange
- **OLS**: Ordinary Least Squares
- **QLIKE**: Quasi-LIKElihood (loss function)
- **RF**: Random Forest
- **RK**: Realised Kernel
- **RQ**: Realised Quarticity
- **RV**: Realised Variance
- **SE**: Standard Error
- **SHAR**: Semivariance HAR model
- **TQ**: Tripower Quarticity
- **TRV**: Truncated Realised Variance
- **TSRV**: Two-Scale Realised Variance
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- **VIX**: CBOE Volatility IndeX
- **VRP**: Variance Risk Premium
- **HAR-X**: Heterogeneous AutoRegressive model with additional regressors
- **DOI**: Digital Object Identifier

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